Online Supplementary Material

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Illustrative Model

This online supplementary material presents a DSGE model with an economically motivated structure to investigate the theoretical findings on identification and estimation under the proposed two-stage block-wise plug-in estimation strategy. Although the DSGE model serves as an illustrative application of the proposed theoretical framework, the supplementary material also provides mathematical proofs for the economically motivated structure, along with the results of DSGE estimation, identification analysis, extensions, and sensitivity analysis. In addition, it includes a policy counterfactual analysis of carbon taxation and a numerical analysis of desert greening.

1. Introduction to an Economically Motivated Structure for Sequential Estimation

The theoretical findings on identification and estimation presented in the main text are illustrated using an economic model in this appendix. Specifically, we develop a structural representation of a fossil-fuel-based, two-sector economy in which electricity demand is derived from manufacturing production and system-wide CO_2 accumulation is mitigated through carbon capture and storage (CCS) and vegetation restoration. The sequential relationship between manufacturing production and electricity demand provides a natural structural setting for applying the block-wise plug-in estimation strategy.

We estimate the DSGE model using U.S. data and examine the empirical and computational relevance of the theoretical findings through identification diagnostics and estimation sensitivity analyses. Given the substantial carbon-removal requirements implied by the analysis, together with the costs and implementation challenges associated with CCS, we further derive policy implications for carbon mitigation strategies through numerical exercises on carbon taxation and vegetation restoration.

The literature on carbon mitigation has primarily focused on carbon taxation (e.g., Heutel, 2012; Golosov et al., 2014; Kotlikoff et al., 2021; Matsumura et al., 2024) and, to a lesser extent, CCS (e.g., Barron et al., 2018). By jointly examining CCS, carbon taxation, and vegetation restoration within a unified DSGE framework, our paper broadens the range of carbon mitigation mechanisms considered in structural macroeconomic analysis while applying the proposed block-wise plug-in strategy for DSGE estimation.

2. Economic Structure and DSGE Model

This section develops the economic structure underlying the theoretical framework presented in Section 3 of the main text. It covers household preferences and constraints, production and factor allocation, capital accumulation and investment, technological dynamics, and the resulting DSGE model. The subsequent subsection directs readers to the relevant appendices of this supplementary material for the DSGE estimation results, identification analysis, diagnostic plots, sensitivity analysis, and extension exercises.

2.1. Household

2.1.1. Welfare

We assume that the representative household seeks to maximize utility, which is increasing in household consumption C_t and decreasing in the economy's CO_2 stock D_t at time t. This setup is represented by the following quasi-linear welfare function, as in Berliant et al. (2014):

$$\max \mathrm{E}_0\left[\int_{t=0}^{\infty} e^{-\rho t}(\ln C_t - D_t)dt\right] \quad (1)$$

The expectation operator $\mathrm{E}_0[.]$ denotes expectations conditional on the information set available at time 0. Uncertainty arises from stochastic disturbances associated with the evolution of the CO_2 stock D_t, which governs carbon-related damages. ρ denotes the subjective discount rate. The specification is related to Heutel (2012), where environmental damages enter household utility in a dynamic optimization framework, but is closer to

Tahvonen (1995), where consumption and environmental damages enter contemporaneously.

Two modeling conventions are worth noting. First, producer welfare is omitted because firms are owned by the representative household and profits are fully rebated. Second, labor supply is not chosen by the household because utility depends only on consumption and the CO_2 stock; population growth therefore affects aggregate labor input rather than fertility.

2.1.2. Budget constraint

Define the representative household's budget constraint at time t as:

$$w\bar{L}_t + r\bar{K}_t + \Pi_t - S_t - pC_t - \phi_D D_t = 0 \quad (2)$$

Here, w and r denote the wage rate and return to capital; $\bar{L}_t$ and $\bar{K}_t$ denote labor (number of workers) and physical capital. ϕ_D and p are the unit carbon-damage cost and manufacturing-goods price, respectively. Saving S_t finances capital accumulation depicted in Equation (8). Π_t denotes the lump-sum profit rebate to the representative household from the manufacturing and electricity sectors. The rebate is not necessarily equal to the two sectors' contemporaneous total profits, which vary over time with their pricing markups.

Equation (2) follows Heutel (2012) and Golosov et al. (2014) but differs in two respects. First, the household owns the savings that directly finance investment in the manufacturing and electricity sectors. Second, CO_2 damages enter the model through two distinct channels. The term D_t in the household's utility function captures the direct disutility associated with environmental degradation, reflecting the global nature of emissions and the incomplete internalization of the associated carbon externality. Meanwhile, $\phi_D D_t$ in the budget constraint represents the associated monetary cost, which reduces resources available for consumption. These two terms capture distinct effects of the same environmental externality rather than duplicating the same loss.

Hereafter, the time subscript t is omitted for readability unless needed.

2.2. Manufacturing and electricity sectors

Both the manufacturing and electricity sectors operate under imperfect competition. While standard in manufacturing, this assumption is consistent with the partly regulated U.S. electricity market, where regulated monopolies coexist with competitive markets (EPA, 2023b). In 2017, the U.S. had 2,938 electric utilities, comprising cooperatives, publicly owned, and investor-owned utilities (Lindstrom and Hoff, 2019). Hereafter, superscripts Y and E denote the manufacturing and electricity sectors, respectively.

2.2.1. Manufacturing

Electricity, fossil fuels, labor, and physical capital used in producing manufacturing output Y are denoted by E^Y, F^Y, L^Y, and K^Y, respectively. Fossil fuels serve as production inputs, while electricity is supplied by the electricity sector. E^Y should not be confused with the expectation operator in Section 2.1.1. The unit costs associated with these factors are represented by e, f, w, and r, respectively. In equilibrium, production cost is assumed to be a fraction m_Y^{-1} of revenue, where the revenue-to-cost ratio $m_Y > 1$ captures markups or other wedges between price and average cost. For simplicity, we refer to m_Y as the manufacturing sector's price markup. The resulting production cost is therefore expressed as

$$pm_Y^{-1}Y = eE^Y + fF^Y + wL^Y + rK^Y. \tag{3}$$

In **Appendix A2.1**, we relate the optimal input choices in (3) to marginal products and factor prices through the first-order conditions, or to marginal revenue products under imperfect competition. The resulting input–output coefficients are equilibrium objects that vary with factor prices and production conditions, rather than fixed technological parameters, and, as shown in Remark 1, determine cross-sectoral factor allocations.

2.2.2. Electricity generation

All electricity produced by the electricity sector is used as an input in manufacturing, so

electricity supply equals manufacturing demand. Let $m_E > 1$ denote the sector's revenue-to-cost ratio, which reflects its markup or other price–cost wedge. For simplicity, we refer to m_E as the markup. Fossil fuels enter electricity generation through the fossil fuel–output ratio b_{EF}, alongside labor and physical capital, using the same notation and unit cost definitions as in the manufacturing sector. Thus, equilibrium costs satisfy:

$$em_E^{-1}E^Y = fb_{EF}E^Y + wL^E + rK^E, \tag{4}$$

where $b_{EF}E^Y := F^E$ denotes the amount of fossil fuel used in the electricity sector. As in Equation (3), optimal input choices depend on input prices and marginal revenue. Non-fossil energy sources are allowed, with negligible input and installation costs incorporated into physical capital K^E and its rental cost.[1] Electricity generation unexplained by E^Y, L^E, and K^E is attributed to technological progress associated with the use of alternative energy sources.

2.2.3. Factor allocations in the manufacturing and electricity sectors

Given the total amounts of fossil fuels $\bar{F}$, labor $\bar{L}$, and physical capital $\bar{K}$ allocated between the manufacturing and electricity sectors, the corresponding resource constraints are:

$$F^Y + F^E = \bar{F}, \tag{5}$$

$$L^Y + L^E = \bar{L}, \tag{6}$$

$$K^Y + K^E = \bar{K}, \tag{7}$$

As noted earlier, $F^E = b_{EF}E^Y$. We further define $a_{Yi} = \frac{i^Y}{Y}$ for $i = F, K, L, E$, and let the superscript * denote the instantaneous growth rate. Then, using Equations (3)–(7), we obtain the following endogenous time-varying input–output coefficients:

Remark 1. *For* $i = F, K, L$, $b_{Ei}^* \approx -a_{Yi}^*$ *when* $a_{Yi} \approx a_{YE}b_{Ei}$ *holds in an equilibrium*

[1] We abstract from marginal costs of alternative energy sources due to the low marginal fuel costs of nuclear power (World Nuclear Association, 2023) and the negligible marginal energy input costs of wind and solar.

characterized by (i) $E^{Y*} = \bar{F}^* = \bar{L}^* = \bar{K}^*$ *and (ii)* $Y^* - E^{Y*} \in [0, \zeta(\bar{L})]$, *where* $\zeta(\bar{L}) \in (0,1)$ *and* $\lim_{\bar{L}\to\infty} \bar{L}\zeta(\bar{L}) = 0$.

Proof: See **Appendix A1.1**.

The condition $E^{Y*} = \bar{F}^* = \bar{L}^* = \bar{K}^*$ imposes a common proportional growth-rate restriction across factor-augmented inputs. Under $0 \le Y^* - E^{Y*} \le \zeta(\bar{L})$ and $\lim_{\bar{L}\to\infty} \bar{L}\zeta(\bar{L}) = 0$, the growth-rate differential between manufacturing output and electricity production vanishes asymptotically, and the condition $\bar{L}\zeta(\bar{L}) \to 0$ also ensures that the associated input-allocation remainder is asymptotically negligible. Remark 1 therefore shows that, under these conditions, changes in the direct input–output coefficient in manufacturing are offset by corresponding changes in the input requirement induced through electricity production when the direct and electricity-mediated input requirements satisfy $a_{Yi} \approx a_{YE} b_{Ei}$.

2.3. Capital accumulation and investment requirements

Investment goods, produced by the manufacturing sector, include office equipment (e.g., printers and computers) and industrial tools (e.g., welding equipment) (Segal, 2020). Let v_K denote the physical investment-to-output ratio, so that investment expenditure is $pv_K Y$. In equilibrium, investment expenditure is financed by household saving, so nominal saving satisfies $S = pv_K Y$, where p is the price of investment goods, assumed to equal the price of consumption goods for simplicity. Capital in the manufacturing and electricity sectors, K^Y and K^E, depreciates at the rate δ. Provided that capital is measured in the same value units as investment expenditure, the total capital stock evolves as

$$\dot{\bar{K}} = pv_K Y - (K^Y + K^E)\delta, \text{ where } pv_K Y = S. \tag{8}$$

We can derive an expression for the investment-to-output ratio v_K from (8). Define $a_{YK} = \frac{K^Y}{Y}$ and $b_{EK} = \frac{K^E}{E^Y}$ as the capital-output ratios in manufacturing and electricity,

respectively, $a_{YE} = \frac{E^Y}{Y}$ as the electricity intensity of manufacturing, and $\kappa_K \equiv \frac{a_{YK}^* + Y^* + \delta}{p}$ and $\kappa_E \equiv \frac{b_{EK}^* + E^{Y*} + \delta}{p}$ as the corresponding effective capital-replenishment coefficients, where * denotes the instantaneous growth rate. The resulting expression is summarized in Remark 2:

Remark 2. *The investment-to-output ratio* v_K *is given by* $v_K = \kappa_K a_{YK} + \kappa_E a_{YE} b_{EK}$*, where* $\kappa_K a_{YK}$ *represents the direct contribution of manufacturing capital to investment requirements, while* $\kappa_E a_{YE} b_{EK}$ *represents the indirect contribution arising from the capital-replenishment requirements of electricity production.*

Proof: See **Appendix A1.2**.

2.4. Technology

2.4.1. CO_2 accumulation

The CO_2 stock D increases with manufacturing-related emissions, including direct production emissions and indirect emissions from electricity generation. Let $\tilde{\phi}_F$ denote total emissions per unit of manufacturing output from both sources. Although structurally determined by these sources, **Appendix B2** suggests that $\tilde{\phi}_F$ is sufficiently stable to be treated as constant. The CO_2 stock D declines through three channels: (i) CCS mitigation, which reduces emissions at rate τ_s; (ii) natural absorption at rate τ_{n0}, representing baseline absorption in Figure 1; and (iii) additional absorption from vegetation restoration at rate τ_{nx}.

Thus, as in Plourde (1972) and Tahvonen (1995), the evolution of D is given by

$$\dot{D} = (1 - \tau_s)\tilde{\phi}_F Y - (\tau_{n0} + \tau_{nx})D. \tag{9}$$

This representation captures domestic carbon dynamics, while foreign emissions and other unmodeled sources of variation are treated as stochastic disturbances in the empirical specification rather than modeled explicitly in Equation (9). Although D reduces household welfare, households take it as given, as in Heutel (2012) and Golosov et al. (2014).

2.4.2. R&D of manufacturing technology and electricity generation

R&D funding availability is reflected indirectly through the revenue-to-cost ratios, m_Y and m_E, estimated as market-structure parameters, while technological progress is not explicitly modeled in the main text, following Heutel (2012) and Golosov et al. (2014). This issue is further discussed in **Appendix A2.2** within a semi-endogenous growth framework, where cross-country differences in market structure and initial technology levels are shown to generate differences in equilibrium technology paths.

2.5. The DSGE model

The following equations present the discretized representation of our full DSGE system, where, as in Fukač and Pagan (2010), the system's composite parameters are mappings of underlying primitives and model variables:

$$\Omega_0 = \mathrm{E_t}\left[\frac{C_{t+1}}{C_t}\right] \tag{10}$$

$$C_t = \Omega_1 Y_t - \Omega_2 D_t + \frac{\Pi_t}{p} \tag{11}$$

$$\Delta D = \Omega_3 Y - \Omega_4 D \tag{12}$$

$$v_K = \kappa_K a_{YK} + \kappa_E a_{YE} b_{EK} \tag{13}$$

$$\Omega_1 = \frac{1}{p}\left[\left((m_E^{-1} - 1)e - b_{EF} f\right)a_{YE} + \left((m_Y^{-1} - 1)p - a_{YF} f\right) + (1 - v_K)p\right] \tag{11-1}$$

$$\Omega_2 = \frac{\phi_D}{p} \tag{11-2}$$

$$\Omega_3 = (1 - \tau_s)\tilde{\phi}_F \tag{12-1}$$

$$\Omega_4 = \tau_{n0} + \tau_{nx} \tag{12-2}$$

Here, a_{YF}, defined as $\frac{F^Y}{Y}$, represents the fossil fuel–output ratio in manufacturing. Variables Y, C, e, p, and τ_{n0} are time-varying variables observed in the data, and D, v_K, a_{YF}, a_{YK}, b_{EK}, a_{YE}, and b_{EF} are computed using observed data. Expressing the equilibrium system in terms of input-output coefficients rather than levels abstracts from scale variation, yields a more

compact representation of the equilibrium conditions, and facilitates numerical estimation of the DSGE model.

This empirical DSGE representation is obtained by compressing the theoretically derived equilibrium system rather than by directly specifying and estimating the complete set of primitive first-order conditions. Specifically, Equation (10), derived in **Appendix A1.3**, provides a reduced-form representation of the Euler equation. As shown in the derivation, Ω_0 depends on the net return on capital, the inflation rate, and the household's time preference. Assuming that the equilibrium real return remains constant over the estimation period and that time preference is stable, Ω_0 is treated as a constant reduced-form equilibrium parameter rather than a primitive preference parameter. Equation (11) represents a simplified single-period household budget constraint, with its full derivation provided in **Appendix A1.4**. The coefficient Ω_1 in Equation (11) is a composite equilibrium parameter whose structural mapping is given in Equation (11-1). This mapping combines underlying structural primitives with model variables observed in the data. While the markups m_Y and m_E in Equation (11-1) affect the price of manufacturing goods through their respective production cost structures, the profit payment $\frac{\Pi_t}{p}$ to the representative household is treated as a lump-sum payment and is not required to equal the combined contemporaneous profits of the manufacturing and electricity sectors in every period. This distinction in Equation (11) allows the model to retain the influence of both sectors' markups on output price determination while allowing their combined profits to deviate temporarily from the payment received by the household.

Equation (12) describes the dynamics of the CO_2 stock. It is obtained by discretizing an equation derived from Equation (9) using the forward Euler method, as detailed in **Appendix A1.4**. Equation (13), derived as Equation (A-5') in the proof of Remark 2 in **Appendix A1.2**, links the investment-to-output ratio v_K to the input-output coefficients of capital in the manufacturing and electricity sectors.

Equations (10)–(13), (11-1)–(11-2), and (12-1)–(12-2) jointly complete the equilibrium system. As discussed in Remark 1, factor allocation within the equilibrium system can be characterized in terms of input–output coefficients. Although these coefficients may vary over time, Remark 1 shows that, under diminishing scale effects, the growth-rate differential between manufacturing output and electricity production vanishes asymptotically, while the associated input-allocation remainder becomes asymptotically negligible.

Rather than using physical capital accumulation as the primary state-transition mechanism, our framework emphasizes carbon-stock dynamics, with Equation (12) specifying accumulated CO_2 stock as the key state variable governing intertemporal propagation. Physical capital formation is represented by the investment-to-output ratio in Equation (13), which is determined by the capital-output and electricity-output coefficients. Thus, while capital remains an important production input, it affects the system through the investment relationship rather than through a separate capital accumulation equation.

While the difficulty of estimating the entire DSGE system motivates its block-wise decomposition, the presence of Ω_1 in Equations (11) and (11-1) within the full system presented above motivates its use as a plug-in restriction.

2.6. Decomposition of the DSGE system

The DSGE model presented in Section 2.5 is decomposed into two blocks: the macroeconomic block and the sectoral block. We first introduce the macroeconomic block, followed by the sectoral block, which is conditional on a plug-in restriction obtained from the macroeconomic block.

1. Macroeconomic block

This block comprises Equations (10)–(12). In this block, the parameters Ω_0, Ω_1, Ω_2, Ω_3, and Ω_4 are estimated from the data. Ω_0 is a reduced-form equilibrium parameter rather than

a discount factor, while the economic intuition for the remaining parameters is provided in **Appendix A1.4**.

As noted earlier, before imposing the equilibrium conditions, the total profits generated by the two sectors under markup pricing vary over time and may therefore differ from the lump-sum rebate $\frac{\Pi_t}{p}$. However, **Appendix A2.1** shows that, under the equilibrium conditions of the manufacturing sector, its profit can be expressed as a fixed proportion of aggregate output. The same reasoning applies to the electricity sector under its corresponding equilibrium conditions. These results imply that, in equilibrium, the lump-sum rebate equals the sum of the profits of the two sectors. Accordingly, using these equilibrium relationships, we express $\frac{\Pi_t}{p}$ as a fixed proportion of aggregate output and absorb it into the output coefficient. Equation (11) can therefore be rewritten as:

$$C_t = \Omega_1 Y_t - \Omega_2 D_t \tag{11'}$$

Although the term $\frac{\Pi_t}{p}$ disappears from Equation (11') because total profits are absorbed into the output coefficient, this simplification does not result from directly offsetting the lump-sum profit rebate against the markup-related profit component in Ω_1 in Equation (11). Such an offset imposed before the equilibrium conditions are established would eliminate the markup-related component and, consequently, undermine the role of markups in price determination. By contrast, the present derivation absorbs profits into the output coefficient only after imposing the equilibrium conditions, thereby preserving the role of markups in price determination.

In the DSGE model implementation, Ω_1 is an equilibrium-mapped coefficient whose posterior distribution reflects uncertainty in the underlying structural parameters transmitted through the equilibrium relationships within the Bayesian estimation framework.

Let the state vector be defined as $\boldsymbol{x_t} = [C_t \quad Y_t \quad D_t]^T$. The macroeconomic block can then be expressed as:

$$A_M \mathrm{E}_t \boldsymbol{x_{t+1}} = B_M \boldsymbol{x_t} + \boldsymbol{\epsilon_t}, \tag{14}$$

where $\boldsymbol{\epsilon_t}$ captures structural disturbances and

$$A_M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; B_M = \begin{bmatrix} \Omega_0 & 0 & 0 \\ -1 & \Omega_1 & -\Omega_2 \\ 0 & \Omega_3 & -(\Omega_4 - 1) \end{bmatrix}.$$

2. Sectoral block

This block is represented by Equations (11-1) and (13), where the parameters f, m_Y, m_E, κ_K, and κ_E are estimated. A nonzero value of f implies the continuous use of fossil fuels in equilibrium. The roles of m_Y and m_E are discussed in relation to Equations (3) and (4), while the economic intuition for κ_K and κ_E is provided in the discussion of Remark 2. As discussed in Equation (8), κ_K and κ_E are themselves composite equilibrium parameters that depend on underlying structural primitives, but they are estimated directly at the composite-parameter level rather than through separate estimation of those underlying primitives. In the implementation of the DSGE model, the uncertainty associated with their underlying primitives is assumed to propagate through the corresponding equilibrium mappings and be reflected in their posterior distributions, analogously to Ω_1 in Equation (11').

The plug-in restriction in this block is Ω_1, obtained from the macroeconomic block. As the coefficient capturing the effect of manufacturing output on consumption in Equation (11'), Ω_1 depends on the market structure parameter m_E of the electricity sector, conditional on the market structure parameter m_Y of the manufacturing sector (Equation (11-1)). It provides a structural linkage between the manufacturing and electricity sectors and, consistent with Proposition 1, contracts the admissible parameter region of the sectoral block while preserving the identifying equations associated with m_Y, m_E, f, κ_K, and κ_E.

The preservation of identifying information follows from the nonlinear structure of Equation (11-1). In particular, Equation (11-1) contains at least two remaining free parameters, so replacing Ω_1 with its first-stage estimate does not exhaust the identifying role of this equation. This feature is not specific to Ω_1: any plug-in restriction that enters an equation containing sufficient remaining free parameters can preserve the identifying information for those parameters, as formalized by Proposition 1(ii). The same logic extends to settings with multiple plug-in restrictions and additional estimation blocks, provided that each plug-in restriction satisfies the corresponding preservation condition. Thus, the single plug-in restriction used in this application provides a transparent empirical illustration of the general identification mechanism underlying the proposed framework.

Define $\boldsymbol{y_t} = [v_K \;\; \Omega_1 - (1 - v_K)]^T$ and $\boldsymbol{z_t} = [a_{YK} \;\; a_{YEK} \;\; a_{YE} \;\; a_{YF} \;\; 1]^T$, where $a_{YEK} = a_{YE} b_{EK}$ is a composite term constructed from observable series, whereas b_{EK} itself is not directly observed. Equations (11-1) and (13) can be written as

$$A_S \boldsymbol{y_t} = B_{S,t}(\boldsymbol{\Omega_f}) \boldsymbol{z_t} + \boldsymbol{\epsilon'_t} \tag{15}$$

where $A_S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\boldsymbol{\Omega_f} = (m_E, m_Y, f, \kappa_K, \kappa_E)$, and $\boldsymbol{\epsilon'_t}$ captures structural disturbances. The coefficient matrix is given by $B_{S,t}(\boldsymbol{\Omega_f}) = \begin{bmatrix} \kappa_K & \kappa_E & 0 & 0 & 0 \\ 0 & 0 & \widetilde{\Omega}_{SE} & \widetilde{\Omega}_{SF} & (m_Y^{-1} - 1) \end{bmatrix}$, where $\widetilde{\Omega}_{SE} = \frac{1}{p}[(m_E^{-1} - 1)e - b_{EF} f]$ and $\widetilde{\Omega}_{SF} = -\frac{f}{p}$. The variables p, e, and b_{EF} are observable series entering the construction of composite coefficients but are not state variables of the model.

2.7. Summary of estimation results, diagnostic plots, and supplementary analyses

The estimation results and their economic implications for the macroeconomic and sectoral blocks are presented in **Appendix B3**. The posterior diagnostic plots for the two blocks are provided in **Appendix C1**.

Appendix B4 compares posterior estimates obtained under the baseline specification with those obtained using a larger number of MCMC draws. **Appendix B5** presents an extended-

sample DSGE estimation exercise in which the unavailable pre-1990 CO_2 stock and electricity price series are reconstructed using the estimated parameters reported in Tables B2 and B3 of **Appendix B3**, together with other observed variables.

Appendix D1 presents a policy counterfactual analysis of the effects of carbon taxation on consumption, output, and the CO_2 stock. Furthermore, given the modeled costs of CCS and carbon taxation, **Appendix D2** examines the potential of desert greening as a complementary carbon-sequestration strategy.

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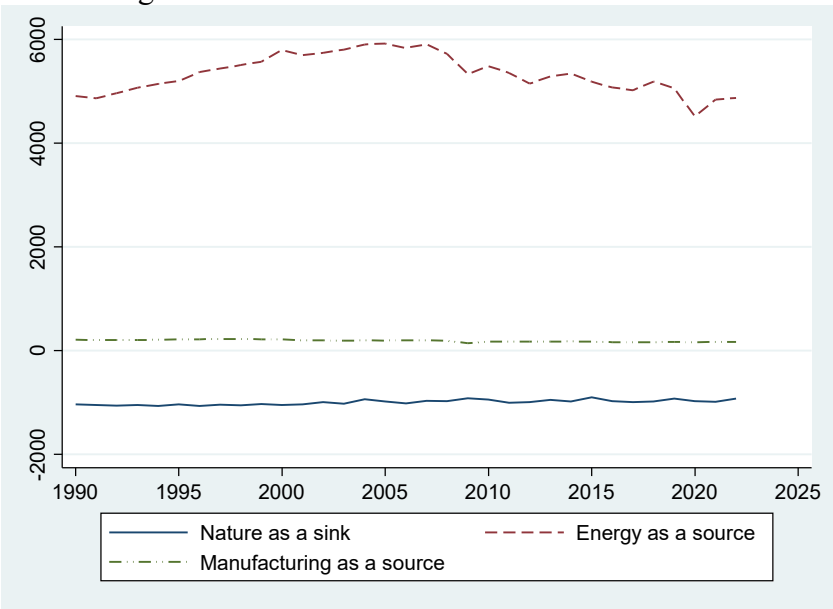
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Figure 1. Carbon emissions in the United States



Notes:

(i) The horizontal axis indicates the year, and the vertical axis indicates the level of CO_2 emissions. The unit on the vertical axis is million metric tons (MMT).

(ii) Carbon emissions due to energy use (as shown by the dashed curve) include emissions from the manufacturing, electricity, transport, residential, and commercial sectors. This differs from emissions arising from the use of fossil fuels as production inputs (shown by the dash–dotted curve).

(iii) In the United States in 2021, greenhouse gas emissions were distributed across sectors as follows: transportation accounted for 28%, the electricity sector for 25%, the industrial sector for 23%, the commercial and residential sector for 13%, and the agricultural sector for 10% (EPA, 2023a). Despite these sectoral weights, it is important to recognize that energy is integral to all sectors for production and household maintenance. This figure provides a direct observation of the significant role of energy across various sectors and manifests the importance of reducing carbon emissions due to energy use while simultaneously enhancing the Earth's carbon absorption capabilities.

(iv) The data are sourced from the U.S. Environmental Protection Agency (2025), *Inventory of U.S. Greenhouse Gas Emissions and Sinks*, available at: https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks

Appendix A: Supplementary Mathematical Derivations and Theoretical Analysis of the Illustrative Model

Appendix A1: Mathematical Derivations and Proofs of the Illustrative Model

A1.1. Proof of Remark 1

Remark 1 explains how producers' profit-maximizing behavior determines the equilibrium allocation of production inputs across sectors. The proof combines Equations (3) and (4), which characterize producers' profit-maximizing behavior, with Equations (5)–(7), which describe the equilibrium allocation of production inputs across sectors:

$$pm_Y^{-1}Y = eE^Y + fF^Y + wL^Y + rK^Y,$$

$$em_E^{-1}E^Y = fF^E + wL^E + rK^E,$$

$$F^Y + F^E = \bar{F},$$

$$L^Y + L^E = \bar{L},$$

$$K^Y + K^E = \bar{K},$$

where $F^E = b_{EF}E^Y$.

The conditions required for this remark are the following assumptions:

Assumption A1-1. In equilibrium, $E^{Y*} = \bar{F}^* = \bar{L}^* = \bar{K}^*$.

Assumption A1-2. In equilibrium, $0 \leq Y^* - E^{Y*} \leq \zeta(\bar{L})$, where $\lim_{\bar{L}\to\infty} \bar{L}\zeta(\bar{L}) = 0$ and $\zeta(\bar{L}) \in (0,1)$.

The relationship $E^{Y*} = \bar{F}^* = \bar{L}^* = \bar{K}^*$ in Assumption A1-1 imposes a common proportional growth-rate restriction across factor-augmented inputs.

The inequality $0 \leq Y^* - E^{Y*} \leq \zeta(\bar{L})$ is assumed to capture the fact that output growth is not driven solely by electricity use. Substituting $E^{Y*} = \bar{L}^*$ from Assumption A1-1 into this inequality yields $0 \leq (Y^* - \bar{L}^*) \leq \zeta(\bar{L})$. Multiplying both sides of the resulting inequality by $\bar{L}$ gives $0 \leq (Y^* - \bar{L}^*)\bar{L} \leq \bar{L}\zeta(\bar{L})$. Under the condition $\lim_{\bar{L}\to\infty} \bar{L}\zeta(\bar{L}) = 0$ (Assumption A1-2), it follows that $(Y^* - \bar{L}^*)\bar{L} \to 0$, implying that the difference between Y^* and E^{Y*} becomes asymptotically negligible as $\bar{L}$ grows. Combined with Assumption A1-1 again, this result

implies that Y^*, $\bar{L}^*$, $\bar{K}^*$, and E^{Y*} converge to approximately the same growth rate in the limit.

To facilitate the characterization of production input allocation, we introduce input–output coefficients for the manufacturing sector: a_{YE}, a_{YF}, a_{YL}, and a_{YK}, defined as the ratios of electricity, fossil fuels, labor, and physical capital to total output Y, respectively. That is, $a_{Yi} = \frac{i^Y}{Y}$ for $i = F, K, L, E$. We also introduce input–output coefficients for the electricity sector: $b_{EL} = \frac{L^E}{E^Y}$, $b_{EK} = \frac{K^E}{E^Y}$, and $b_{EF} = \frac{F^E}{E^Y}$, representing the ratios of labor, capital, and fossil fuels to electricity output, respectively.

We allow both prices and input–output coefficients to vary over time, and employ an asterisk $*$ to denote the proportional growth rate of a variable over a very small time interval, defined as $x^* = \frac{\dot{x}}{x}$. We use this notation for all input–output coefficients, including a_{Yi} and b_{Ei} for production input i. It is also applied to the revenue–cost ratios and factor prices, including m_Y, m_E, p, e, f, r, and w, all of which are allowed to vary over time.

Using these definitions and notational conventions, we now characterize the dynamic implications of producers' profit-maximizing behavior by taking the time derivatives of Equations (3) and (4). The resulting expressions are given below, while the detailed derivations are provided in the **sub-appendix** to this proof:

$$\theta_0 = \theta_E a_{YE}^* + \theta_F a_{YF}^* + \theta_L a_{YL}^* + \theta_K a_{YK}^* \tag{A-1}$$

$$\mu_0 = \mu_F b_{EF}^* + \mu_L b_{EL}^* + \mu_K b_{EK}^* \tag{A-2}$$

Here,

$$\theta_0 = \frac{(p^* - m_Y^*)p}{m_Y} - (a_{YE}ee^* + a_{YF}ff^* + a_{YL}ww^* + a_{YK}rr^*),$$

$$\theta_E = a_{YE}e,\ \theta_F = a_{YF}f,\ \theta_L = a_{YL}w, \text{ and } \theta_K = a_{YK}r; \text{ and}$$

$$\mu_0 = \frac{(e^* - m_E^*)e}{m_E} - (b_{EF}ff^* + b_{EL}ww^* + b_{EK}rr^*),$$

$$\mu_F = b_{EF}f,\ \mu_L = b_{EL}w, \text{ and } \mu_K = b_{EK}r.$$

We also take the total time derivatives of Equations (5)–(7). The resulting expressions are

listed below, while their derivations are provided in the **sub-appendix**:

$$b_{EF}^{*} = \mu_{F0} + \mu_{F1} a_{YF}^{*} \tag{A-3-1}$$

where $\mu_{F0} = \frac{\dot{\bar{F}} - a_{YF} Y^{*} Y - a_{YE} b_{EF} E^{Y*} Y}{a_{YE} b_{EF} Y}$ and $\mu_{F1} = -\frac{a_{YF}}{a_{YE} b_{EF}}$.

$$b_{EL}^{*} = \mu_{L0} + \mu_{L1} a_{YL}^{*} \tag{A-3-2}$$

where $\mu_{L0} = \frac{\dot{\bar{L}} - a_{YL} Y^{*} Y - a_{YE} b_{EL} E^{Y*} Y}{a_{YE} b_{EL} Y}$ and $\mu_{L1} = -\frac{a_{YL}}{a_{YE} b_{EL}}$.

$$b_{EK}^{*} = \mu_{K0} + \mu_{K1} a_{YK}^{*} \tag{A-3-3}$$

where $\mu_{K0} = \frac{\dot{\bar{K}} - a_{YK} Y^{*} Y - a_{YE} b_{EK} E^{Y*} Y}{a_{YE} b_{EK} Y}$ and $\mu_{K1} = -\frac{a_{YK}}{a_{YE} b_{EK}}$.

Using Assumptions A1-1 and A1-2, we further show in the **sub-appendix** of this appendix that, in Equations (A-3-1) –(A-3-3),

$$\dot{\bar{\imath}} \approx a_{Yi} Y^{*} Y + a_{YE} b_{Ei} E^{Y*} Y, \text{ with } \bar{\imath} = i^{Y} + i^{E} \text{ for } i = F, L, K. \tag{A-4}$$

Substituting Equation (A-4) into Equations (A-3-1)–(A-3-3) yields $\mu_{F0} \approx 0$, $\mu_{L0} \approx 0$, and $\mu_{K0} \approx 0$. Thus,

$$b_{Ei}^{*} \approx \mu_{i1} a_{Yi}^{*}, \text{ where } \mu_{i1} = -\frac{a_{Yi}}{a_{YE} b_{Ei}} \text{ and } i = F, L, K.$$

In the above expression, μ_{i1} captures the relative magnitude of direct manufacturing input requirements and electricity-mediated input requirements for factor i. Since a_{Yi}, a_{YE}, and b_{Ei} are positive input–output coefficients, $\mu_{i1} < 0$. Therefore,

Remark 1**:** *For* $i = F, K, L$, $b_{Ei}^{*} \approx -a_{Yi}^{*}$ *when* $a_{Yi} \approx a_{YE} b_{Ei}$ *holds in an equilibrium characterized by (i)* $E^{Y*} = \bar{F}^{*} = \bar{L}^{*} = \bar{K}^{*}$ *and (ii)* $Y^{*} - E^{Y*} \in [0, \zeta(\bar{L})]$, *where* $\zeta(\bar{L}) \in (0,1)$ *and* $\lim_{\bar{L} \to \infty} \bar{L} \zeta(\bar{L}) = 0$.

Q.E.D.

Sub-Appendix to Proof of Remark 1: Derivation of Equations (A-1) and (A-2)

Equations (A-1) and (A-2) are derived from Equations (3) and (4) using the approach pioneered by Jones (1965), which is particularly well-suited for studying the allocation of

production factors across sectors over time.

We define $a^*_{Yi} = \frac{\dot{a}_{Yi}}{a_{Yi}}$ and $b^*_{Ei} = \frac{\dot{b}_{Ei}}{b_{Ei}}$ for $i = L, K$, as well as $a^*_{YE} = \frac{\dot{a}_{YE}}{a_{YE}}$, $a^*_{YF} = \frac{\dot{a}_{YF}}{a_{YF}}$, $b^*_{EF} = \frac{\dot{b}_{EF}}{b_{EF}}$, $m^*_Y = \frac{\dot{m}_Y}{m_Y}$, $m^*_E = \frac{\dot{m}_E}{m_E}$, $p^* = \frac{\dot{p}}{p}$, $e^* = \frac{\dot{e}}{e}$, $f^* = \frac{\dot{f}}{f}$, $r^* = \frac{\dot{r}}{r}$, and $w^* = \frac{\dot{w}}{w}$.

On this basis, we restate Equation (3) as follows:

$$pm_Y^{-1} - a_{YE}e = a_{YF}f + a_{YL}w + a_{YK}r$$

Taking the total time derivative of this equation yields

$$(\dot{p}m_Y^{-1} - pm_Y^{-2}\,\dot{m}_Y) - \dot{a}_{YE}e - a_{YE}\dot{e} =$$

$$\dot{a}_{YF}f + a_{YF}\dot{f} + \dot{a}_{YL}w + a_{YL}\dot{w} + \dot{a}_{YK}r + a_{YK}\dot{r}.$$

In this equation, $(\dot{p}m_Y^{-1} - pm_Y^{-2}\,\dot{m}_Y)$ can be rewritten as $(p^{-1}\dot{p} - m_Y^{-1}\,\dot{m}_Y)pm_Y^{-1}$. Thus,

$$(p^{-1}\dot{p} - m_Y^{-1}\,\dot{m}_Y)pm_Y^{-1} - \left(\frac{\dot{a}_{YE}e}{a_{YE}e} + \frac{a_{YE}\dot{e}}{a_{YE}e}\right)a_{YE}e$$

$$= \dot{a}_{YF}f + a_{YF}\dot{f} + \dot{a}_{YL}w + a_{YL}\dot{w} + \dot{a}_{YK}r + a_{YK}\dot{r}.$$

The above equation can be further simplified to

$$(p^* - m^*_Y)\,pm_Y^{-1} - (a^*_{YE} + e^*)a_{YE}e$$

$$= (a^*_{YF} + f^*)a_{YF}f + (a^*_{YL} + w^*)a_{YL}w + (a^*_{YK} + r^*)a_{YK}r.$$

Define

$$\theta_0 = \frac{p^* - m^*_Y}{m_Y}\,p - (a_{YE}ee^* + a_{YF}ff^* + a_{YL}ww^* + a_{YK}rr^*),$$

$$\theta_E = a_{YE}e,\ \theta_F = a_{YF}f,\ \theta_L = a_{YL}w,\ \theta_K = a_{YK}r.$$

Thus,

$$\theta_0 = \theta_E a^*_{YE} + \theta_F a^*_{YF} + \theta_L a^*_{YL} + \theta_K a^*_{YK}.$$

The same procedure applies to Equation (4). Dividing both sides by E^Y yields:

$$em_E^{-1} = b_{EF}f + b_{EL}w + b_{EK}r$$

Differentiating both sides with respect to time yields:

$$\dot{e}m_E^{-1} - em_E^{-2}\dot{m}_E$$

$$= (\dot{b}_{EF} f + b_{EF} \dot{f}) + (\dot{b}_{EL} w + b_{EL} \dot{w}) + (\dot{b}_{EK} r + b_{EK} \dot{r}).$$

The left-hand side of this equation can be written as:

$$\dot{e} m_E^{-1} - e m_E^{-2} \dot{m}_E = (e^{-1} \dot{e} - m_E^{-1} \dot{m}_E) e m_E^{-1}$$

$$= (e^* - m_E^*) e m_E^{-1}$$

Thus,

$$(e^* - m_E^*) e m_E^{-1}$$

$$= (b_{EF}^* + f^*) b_{EF} f + (b_{EL}^* + w^*) b_{EL} w + (b_{EK}^* + r^*) b_{EK} r.$$

$$\Rightarrow (e^* - m_E^*) e m_E^{-1} - (b_{EF} f f^* + b_{EL} w w^* + b_{EK} r r^*)$$

$$= f b_{EF} b_{EF}^* + w b_{EL} b_{EL}^* + r b_{EK} b_{EK}^*.$$

Define

$$\mu_0 = \frac{(e^* - m_E^*) e}{m_E} - (b_{EF} f f^* + b_{EL} w w^* + b_{EK} r r^*),$$

$$\mu_F = b_{EF} f,\ \mu_L = b_{EL} w,\ \mu_K = b_{EK} r.$$

Thus,

$$\mu_0 = \mu_F b_{EF}^* + \mu_L b_{EL}^* + \mu_K b_{EK}^*.$$

Q.E.D.

Sub-Appendix to Proof of Remark 1: Derivation of Equations (A-3-1)–(A-3-3)

Start from the identity

$$a_{YF} Y + b_{EF} E^Y = \bar{F},$$

where $a_{YF} Y = F^Y$ and $b_{EF} E^Y = F^E$.

Differentiate with respect to time:

$$\left(\dot{a}_{YF} + a_{YF} \frac{\dot{Y}}{Y}\right) Y + \left(\dot{b}_{EF} + b_{EF} \frac{\dot{E}^Y}{E^Y}\right) E^Y = \dot{\bar{F}}$$

$$\Rightarrow \left(\frac{\dot{a}_{YF}}{a_{YF}} + \frac{\dot{Y}}{Y}\right) a_{YF} Y + \left(\frac{\dot{b}_{EF}}{b_{EF}} + \frac{\dot{E}^Y}{E^Y}\right) b_{EF} \left(\frac{E^Y}{Y}\right) Y = \dot{\bar{F}}$$

where $\dot{\bar{F}}$ denotes the instantaneous rate of change in total fossil-fuel availability.

As defined in the main text, $a_{YE} = \frac{E^Y}{Y}$. A superscript * denotes the instantaneous growth rate. Then, the above equation becomes:

$$(a_{YF}^* + Y^*)a_{YF}Y + (b_{EF}^* + E^{Y*})a_{YE}b_{EF}Y = \dot{\bar{F}}$$

$$\Rightarrow \dot{\bar{F}} - a_{YF}Y^*Y - a_{YE}b_{EF}E^{Y*}Y = a_{YF}a_{YF}^*Y + a_{YE}b_{EF}b_{EF}^*Y$$

$$\Rightarrow b_{EF}^* = (\frac{\dot{\bar{F}} - a_{YF}Y^*Y - a_{YE}b_{EF}E^{Y*}Y}{a_{YE}b_{EF}Y}) - (\frac{a_{YF}}{a_{YE}b_{EF}})a_{YF}^*$$

Define $\mu_{F0} = \frac{\dot{\bar{F}} - a_{YF}Y^*Y - a_{YE}b_{EF}E^{Y*}Y}{a_{YE}b_{EF}Y}$ and $\mu_{F1} = -(\frac{a_{YF}}{a_{YE}b_{EF}})$. Thus, the above expression of b_{EF}^* can be written as:

$$b_{EF}^* = \mu_{F0} + \mu_{F1}a_{YF}^*$$

Applying the same procedure to $a_{YL}Y + b_{EL}E^Y = \bar{L}$ yields

$$b_{EL}^* = \mu_{L0} + \mu_{L1}a_{YL}^*$$

where $\mu_{L0} = (\frac{\dot{\bar{L}} - a_{YL}Y^*Y - a_{YE}b_{EL}E^{Y*}Y}{a_{YE}b_{EL}Y})$ and $\mu_{L1} = -(\frac{a_{YL}}{a_{YE}b_{EL}})$.

From $a_{YK}Y + b_{EK}E^Y = \bar{K}$, we have that

$$b_{EK}^* = \mu_{K0} + \mu_{K1}a_{YK}^*$$

where $\mu_{K0} = (\frac{\dot{\bar{K}} - a_{YK}Y^*Y - a_{YE}b_{EK}E^{Y*}Y}{a_{YE}b_{EK}Y})$ and $\mu_{K1} = -(\frac{a_{YK}}{a_{YE}b_{EK}})$.

Q.E.D.

Sub-Appendix to Proof of Remark 1: Derivation of Equation (A-4)

We decompose the evolution of labor demand into two components: one driven by output growth in the manufacturing sector, and the other reflecting increased labor use in the electricity sector associated with an expansion in economic scale, proxied by labor growth. Let L^Y and L^E denote labor used in manufacturing and electricity generation, respectively. We define:

$$a_{YL} = \frac{L^Y}{Y}$$

$$b_{EL} = \frac{L^E}{E^Y}$$

$$a_{YE} = \frac{E^Y}{Y}$$

$$\bar{L} = L^Y + L^E$$

$$Y^* \equiv \frac{\dot{Y}}{Y}$$

$$E^{Y*} \equiv \frac{\dot{E}^Y}{E^Y}$$

$$\bar{L}^* \equiv \frac{\dot{\bar{L}}}{\bar{L}}$$

On this basis, we express the evolution of labor driven by output growth in the manufacturing sector as $a_{YL}Y^*Y$ and the evolution of labor reflecting increased labor use in the electricity sector associated with an expansion in economic scale as $a_{YE}b_{EL}E^{Y*}Y$. This yields:

$$a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y$$

This expression can be further simplified. The derivation proceeds as follows:

$$a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y$$

$$= \left(\frac{L^Y}{Y}Y^* + \frac{E^Y}{Y}\frac{L^E}{E^Y}E^{Y*}\right)Y \text{ (given that } a_{YL} = \frac{L^Y}{Y},\ a_{YE} = \frac{E^Y}{Y},\ b_{EL} = \frac{L^E}{E^Y}, \text{ and } Y^* = \frac{\dot{Y}}{Y})$$

$$= Y^*L^Y + E^{Y*}L^E$$

$$= L^YY^* + (\bar{L} - L^Y)E^{Y*}$$

$$= L^YY^* + L^EE^{Y*}$$

$$= L^Y(Y^* - E^{Y*}) + \bar{L}E^{Y*}$$

Given $E^{Y*} = \bar{L}^*$ under Assumption A1-1, we have:

$$a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y$$

$$= L^Y(Y^* - \bar{L}^*) + \bar{L}\bar{L}^*$$

$$= L^Y(Y^* - \bar{L}^*) + \dot{\bar{L}}$$

The deviation $Y^* - \bar{L}^*$ in the above expression satisfies $0 \le Y^* - \bar{L}^* \le \zeta(\bar{L})$, which follows from the assumption $0 \le Y^* - E^{Y*} \le \zeta(\bar{L})$ (Assumption A1-2) together with the

growth-rate equalization condition $E^{Y*} = \bar{L}^*$ (Assumption A1-1). Therefore, the expression above satisfies:

$$\dot{\bar{L}} \leq a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y \leq L^Y\zeta(\bar{L}) + \dot{\bar{L}} \leq \bar{L}\zeta(\bar{L}) + \dot{\bar{L}}$$

Since $\lim_{\bar{L}\to\infty} \bar{L}\zeta(\bar{L}) = 0$ (Assumption A1-2), the difference between the upper and lower bounds converges to zero. Hence,

$$a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y = \dot{\bar{L}} + o(1),$$

or equivalently,

$$a_{YL}Y^*Y + a_{YE}b_{EL}E^{Y*}Y \approx \dot{\bar{L}}.$$

Applying the above procedure to the remaining inputs in final goods production yields

$$\dot{\bar{\imath}} \approx a_{Yi}Y^*Y + a_{YE}b_{Ei}E^{Y*}Y, \text{ with } \bar{\imath} = i^Y + i^E \text{ for } i = F, L, K.$$

Q.E.D.

References for Appendix A1.1

Jones, Ronald W. (1965) The structure of simple general equilibrium models. *Journal of Political Economy* 73(6), 557-572.

A1.2. Proof of Remark 2

Using Equation (8), the time derivative of $\overline{K} = a_{YK}Y + b_{EK}E^{Y}$, we derive the following equation, which is shown in the **sub-appendix** of this proof:

$$v_K = \left(\frac{a^*_{YK}+Y^*+\delta}{p}\right) a_{YK} + \left(\frac{b^*_{EK}+E^{Y*}+\delta}{p}\right) a_{YE} b_{EK} \tag{A-5}$$

In this equation, term $(a^*_{YK} + Y^* + \delta)$ combines the instantaneous growth rate of manufacturing capital, $(a^*_{YK} + Y^*)$, with the depreciation rate δ. The term $(b^*_{EK} + E^{Y*} + \delta)$ captures the same fact in the electricity sector.

We regard $\frac{a^*_{YK}+Y^*+\delta}{p}$ as the effective capital-replenishment coefficient of manufacturing production and $\frac{b^*_{EK}+E^{Y*}+\delta}{p}$ as the effective capital-replenishment coefficient of electricity generation. Denote them as κ_K and κ_E, respectively. The above equation is rewritten as follows:

$$v_K = \kappa_K a_{YK} + \kappa_E a_{YE} b_{EK} \tag{A-5'}$$

Q.E.D.

Sub-Appendix to Proof of Remark 2: Derivation of Equation (A-5)

As noted in the proof of Remark 1, $a^*_{YK} = \frac{\dot{a}_{YK}}{a_{YK}}$, $b^*_{EK} = \frac{\dot{b}_{EK}}{b_{EK}}$, $E^{Y*} = \frac{\dot{E}^{Y}}{E^{Y}}$, and $Y^* = \frac{\dot{Y}}{Y}$. Taking time derivatives of $\overline{K} = a_{YK}Y + b_{EK}E^{Y}$ yields:

$$\dot{\overline{K}} = \dot{a}_{YK}Y + a_{YK}\dot{Y} + \dot{b}_{EK}E^{Y} + b_{EK}\,\dot{E^{Y}}$$

$$= (a^*_{YK} + Y^*)a_{YK}Y + (b^*_{EK} + E^{Y*})b_{EK}E^{Y},$$

Substituting $\overline{K} = a_{YK}Y + b_{EK}E^{Y}$ into it gives that:

$$\dot{\overline{K}} = (a^*_{YK} + Y^*)(\overline{K} - b_{EK}E^{Y}) + (b^*_{EK} + E^{Y*})b_{EK}E^{Y}$$

$$= (a^*_{YK} + Y^*)\overline{K} + (b^*_{EK} - a^*_{YK} + E^{Y*} - Y^*)b_{EK}E^{Y}$$

Equating Equation (8) with the expression above for $\dot{\overline{K}}$ yields:

$$pv_K Y - (a_{YK}Y + b_{EK}E^{Y})\delta = (a^*_{YK} + Y^*)\overline{K} + (b^*_{EK} - a^*_{YK} + E^{Y*} - Y^*)b_{EK}E^{Y}.$$

Moreover, because $\overline{K} = a_{YK}Y + b_{EK}E^Y$, the above equation is rewritten as

$$pv_K Y - (a_{YK}Y + b_{EK}E^Y)\delta$$

$$= (a_{YK}^* + Y^*)(a_{YK}Y + b_{EK}E^Y) + (b_{EK}^* - a_{YK}^* + E^{Y*} - Y^*)b_{EK}E^Y$$

$$\Rightarrow [pv_K - \delta a_{YK} - (a_{YK}^* + Y^*)a_{YK}]Y = (b_{EK}^* + E^{Y*} + \delta)b_{EK}E^Y$$

Since $Y > 0$,

$$pv_K - \delta a_{YK} - (a_{YK}^* + Y^*)a_{YK} = (b_{EK}^* + E^{Y*} + \delta)(\frac{b_{EK}E^Y}{Y}).$$

Using $a_{YE} = \frac{E^Y}{Y}$, we obtain:

$$pv_K = (a_{YK}^* + Y^* + \delta)a_{YK} + (b_{EK}^* + E^{Y*} + \delta)a_{YE}b_{EK}$$

That is,

$$v_K = \left(\frac{a_{YK}^* + Y^* + \delta}{p}\right)a_{YK} + \left(\frac{b_{EK}^* + E^{Y*} + \delta}{p}\right)a_{YE}b_{EK}.$$

Q.E.D.

A1.3. Derivation of Equation (10).

In Equation (2), we define the household's flow budget constraint as:

$$w\bar{L}_t + r\overline{K}_t + \Pi_t - S_t - pC_t - \phi_D D_t = 0$$

As noted in the proof of Remark 1, we denote the capital–output coefficients in the manufacturing and electricity sectors by a_{YK} and b_{EK}. The amounts of capital used in the two sectors are given by $a_{YK}Y_t = K_t^Y$ and $b_{EK}E_t^Y = K_t^E$, respectively.

Since $\overline{K}_t = K_t^Y + K_t^E$, we rewrite the law of motion in Equation (8) as

$$\dot{\overline{K}}_t = S_t - \overline{K}_t\delta.$$

Rearranging this law of motion to express S_t and substituting the resulting expression into the representative household's budget constraint in Equation (2) yields:

$$\dot{\overline{K}}_t = w\bar{L}_t + r\overline{K}_t + \Pi_t - pC_t - \phi_D D_t - \overline{K}_t\delta$$

The household maximizes the following expected lifetime utility subject to the above law of motion:

$$\max \mathrm{E}_0\left[\int_{t=0}^{\infty} e^{-\rho t}(\ln C_t - D_t)dt\right]$$

We define the current-value Hamiltonian of this optimization problem as

$$H(t) = \ln C_t - D_t + \lambda_t[w\bar{L}_t + r\overline{K}_t + \Pi_t - pC_t - \phi_D D_t - \overline{K}_t\delta]$$

where λ_t is the costate variable, representing the shadow price of wealth.

The first-order condition with respect to C_t is:

$$\frac{dH(t)}{dC_t} = 0:\ \lambda_t = \frac{1}{pC_t}$$

which implies the following expression by taking logarithms and differentiating with respect to time:

$$\frac{\dot{\lambda}_t}{\lambda_t} = -\frac{\dot{C}_t}{C_t} - \frac{\dot{p}}{p}$$

Provided that the lump-sum profit rebate to the household is invariant to the economy's total capital stock, that is, $\frac{\partial \Pi_t}{\partial \bar{K}_t} = 0$, the costate equation is

$$\dot{\lambda}_t = \rho\lambda_t - \frac{dH(t)}{d\bar{K}_t} = \big(\rho - (r - \delta)\big)\lambda_t.$$

We can further express the above equation as:

$$\frac{\dot{\lambda}_t}{\lambda_t} = \rho - (r - \delta)$$

Combining the above expressions from the FOC and the costate equation yields the Euler equation:

$$\frac{\dot{C}_t}{C_t} = (r - \delta) - \rho - \frac{\dot{p}}{p}$$

The term $r - \delta$ represents the net return on capital facing the household. We assume that this net return, the subjective discount rate ρ, and the inflation rate $\frac{\dot{p}}{p}$ remain constant over a unit time interval. Under these assumptions, consumption growth in discrete time can be expressed as:

$$\frac{C_{t+1}}{C_t} = e^{r-\delta-\rho-\frac{\dot{p}}{p}}$$

Define $\Omega_0 \equiv e^{r-\delta-\rho-\frac{\dot{p}}{p}}$. The corresponding empirical moment condition is therefore

$$\mathrm{E_t}\left[\frac{C_{t+1}}{C_t}\right] = \Omega_0.$$

This representation reflects intertemporal consumption behavior. In the empirical implementation, Ω_0 is estimated as a composite equilibrium parameter rather than as a discount factor.

Q.E.D.

A1.4. Derivations and Interpretations of Equations (11) and (12)

In this section, we derive Equation (11) from the household's budget constraint using Equations (3) and (4), and then apply the same procedure to obtain Equation (12). The economic intuition for the underlying parameters in both equations is also provided within this framework.

Derivation of Equation (11)

Let the inputs in the manufacturing sector—electricity, fossil fuels, labor, and physical capital—be denoted by E^Y, F^Y, L^Y, and K^Y, respectively. These inputs satisfy $E^Y = a_{YE}Y$, $F^Y = a_{YF}Y$, $L^Y = a_{YL}Y$, and $K^Y = a_{YK}Y$. Electricity is assumed to be used exclusively by the manufacturing sector. Using these definitions, Equation (3) can be rewritten as follows:

$$pm_Y^{-1} = a_{YE}e + a_{YF}f + a_{YL}w + a_{YK}r$$

$$\Rightarrow pYm_Y^{-1} - a_{YE}Ye = pYm_Y^{-1} - eE^Y = (a_{YF}Y)f + (a_{YL}Y)w + (a_{YK}Y)r$$

That is,

$$pYm_Y^{-1} = eE^Y + fF^Y + wL^Y + rK^Y,$$

Similarly, we rewrite Equation (4) as follows using $F^E = b_{EF}E^Y$, $L^E = b_{EL}E^Y$, and $K^E = b_{EK}E^Y$:

$$em_E^{-1} = b_{EF}f + b_{EL}w + b_{EK}r$$

$$\Rightarrow\ em_E^{-1} = \frac{F^E}{E^Y}f + \frac{L^E}{E^Y}w + \frac{K^E}{E^Y}r$$

Equivalently,

$$eE^Y m_E^{-1} = fF^E + wL^E + rK^E.$$

Adding the two expressions of pYm_Y^{-1} and $eE^Y m_E^{-1}$ yields:

$$pYm_Y^{-1} + eE^Y m_E^{-1} = eE^Y + f(F^Y + F^E) + w(L^Y + L^E) + r(K^Y + K^E).$$

Let $\bar{F}$, $\bar{L}$, and $\bar{K}$ denote the aggregate endowments allocated between the manufacturing and electricity sectors. Using $F^Y + F^E = \bar{F}$, $L^Y + L^E = \bar{L}$, and $K^Y + K^E = \bar{K}$, the above equation simplifies to:

$$pYm_Y^{-1} + eE^Y m_E^{-1} = eE^Y + f\bar{F} + w\bar{L} + r\bar{K}, \text{ i.e.,}$$

$$w\bar{L} + r\bar{K} = (m_E^{-1} - 1)eE^Y + pYm_Y^{-1} - f(F^E + F^Y)$$

Substituting this expression into the household budget constraint (Equation (2)),

$$w\bar{L}_t + r\bar{K}_t + \Pi_t - pv_K Y_t - pC_t - \phi_D D_t = 0, \text{ with } pv_K Y_t = S_t,$$

yields:

$$(m_E^{-1} - 1)eE_t^Y + (m_Y^{-1} - v_K)pY_t - f(F_t^E + F_t^Y) + \Pi_t - pC_t - \phi_D D_t = 0.$$

Using $F_t^Y = a_{YF}Y_t$, $F_t^E = b_{EF}E_t^Y$, and $E_t^Y = a_{YE}Y_t$, the above equation can be further reduced as follows:

$$(m_E^{-1} - 1)ea_{YE}Y_t + (m_Y^{-1} - v_K)pY_t - f(b_{EF}E_t^Y + a_{YF}Y_t) + \Pi_t - pC_t - \phi_D D_t = 0$$

$$\Rightarrow [(m_E^{-1} - 1)ea_{YE} + (m_Y^{-1} - v_K)p - (b_{EF}a_{YE} + a_{YF})f]Y_t + \Pi_t - pC_t - \phi_D D_t = 0$$

$$\Rightarrow C_t = \frac{1}{p}\left[\left((m_E^{-1} - 1)e - b_{EF}f\right)a_{YE} + \left((m_Y^{-1} - 1)p - a_{YF}f\right) + (1 - v_K)p\right]Y_t - \frac{\phi_D}{p}D_t + \frac{\Pi_t}{p}$$

The markups m_Y and m_E can give rise to profits, but their roles in determining output prices are retained rather than eliminated through substitution into $\frac{\Pi_t}{p}$. Here, $\frac{\Pi_t}{p}$ is treated as a lump-sum profit payment to the representative household and is not assumed to equal the total contemporaneous profits of the manufacturing and electricity sectors in every period. This specification preserves the roles of the two sectors in determining the price of manufacturing goods while allowing their total profits to deviate temporarily from the lump-sum payment received by the household.

Therefore,

$$C_t = \Omega_1 Y_t - \Omega_2 D_t + \frac{\Pi_t}{p}$$

where

$$\Omega_1 = \frac{1}{p}\left[\left((m_E^{-1} - 1)e - b_{EF}f\right)a_{YE} + \left((m_Y^{-1} - 1)p - a_{YF}f\right) + (1 - v_K)p\right]$$

$$\Omega_2 = \frac{\phi_D}{p}.$$

This completes the proof of Equations (11), (11-1), and (11-2).

Q.E.D.

Economic interpretations of Ω_1 and Ω_2

The household budget constraint in Equation (11) accounts for household expenditure on manufacturing net of carbon damage. It consists of two components, $\Omega_1 Y$ and $-\Omega_2 D$. Ω_1 captures the effective amount of household consumption supported by one unit of manufacturing output after accounting for operating profits, factor payments, fossil-fuel costs, and the capital requirement, while Ω_2 measures the marginal impact of carbon damage on consumption.

In Equation (11-1), Ω_1 is expressed as a function of $(m_E^{-1} - 1)e$ and $(m_Y^{-1} - 1)p$. The terms $(1 - m_E^{-1})e$ and $(1 - m_Y^{-1})p$ represent operating profit per unit of electricity output and manufacturing output, respectively. Their negative counterparts enter Ω_1 because factor payments are expressed as revenues net of operating profits.

In addition, Ω_1 is also affected by $a_{YE}b_{EF}f$ and $a_{YF}f$. We treat $a_{YE}b_{EF}f$ and $a_{YF}f$ as production-related costs arising from the use of fossil fuels in both the electricity and manufacturing sectors. As a consequence, consumption is simultaneously linked to fossil fuel dependence through Ω_1.

Equation (11-2) shows that $\Omega_2 = \frac{\phi_D}{p}$, which equals $-\frac{\partial C}{\partial D}$; thus, Ω_2 measures the reduction in consumption associated with a marginal increase in the carbon stock, expressed in units of the consumption good. This formulation links carbon damage—an externality that is difficult to incorporate into producers' profit functions—to household consumption in Equation (11).

Derivation of Equation (12)

Equation (9) is listed as follows:

$$\dot{D} = (1 - \tau_s)\tilde{\phi}_F Y - (\tau_{n0} + \tau_{nx})D$$

The theoretical model is formulated in continuous time to characterize the structural dynamics of consumption, capital accumulation, and the CO_2 stock. For empirical implementation, the continuous-time system is discretized using the forward Euler method. Accordingly, defining $(1 - \tau_s)\tilde{\phi}_F$ and $(\tau_{n0} + \tau_{nx})$ as Ω_3 and Ω_4, respectively, the discretized equation is written as:

$$\Delta D = \Omega_3 Y - \Omega_4 D$$

Economic interpretations of Ω_3 and Ω_4

The evolution of carbon dioxide is depicted by Equation (12). It models the accumulation of the CO_2 stock as a function of higher output (i.e., $\Omega_3 Y$), while its reduction (i.e., $\Omega_4 D$) captures CO_2 removal through natural absorption and human-led vegetation restoration.

Ω_3 captures the effective carbon-emission intensity of manufacturing output after accounting for the carbon-related policy parameter τ_s. Emissions arise either directly from fossil-fuel use in manufacturing or indirectly from fossil-fuel use in electricity generation used by manufacturing.

Ω_4 captures the reduction rate of CO_2. As indicated in Equation (9), this reduction arises from either natural absorption or human-led vegetation restoration. If vegetation loss associated with deforestation dominates natural absorption and any restoration effect, $\tau_{n0} + \tau_{nx}$ may become negative, leading to $\Omega_4 < 0$.

Appendix A2: Supplementary Theoretical Analysis for the Illustrative Model

A2.1. Discussion of Equation (3)

1. Equilibrium relationships among production inputs

Express the profit of the manufacturing sector (denoted by $\tilde{\pi}$) as:

$$\tilde{\pi} = pY - (eE^Y + fF^Y + wL^Y + rK^Y)$$

According to Equation (3), $\frac{pY}{eE^Y+fF^Y+wL^Y+rK^Y} = m_Y$. Under the firm's profit-maximization problem, and treating m_Y as a constant markup, the first-order conditions imply

$$p\frac{\partial Y}{\partial E^Y} = m_Y e,$$

$$p\frac{\partial Y}{\partial F^Y} = m_Y f,$$

$$p\frac{\partial Y}{\partial L^Y} = m_Y w,$$

$$p\frac{\partial Y}{\partial K^Y} = m_Y r.$$

Taking the ratio of any two first-order conditions yields the corresponding equilibrium relationship between the marginal products of production inputs and their respective input prices.

2. Equilibrium revenue–cost representation

Let the optimal solution to the firm's optimization problem be $(F^{Y,o}, E^{Y,o}, L^{Y,o}, K^{Y,o})$, where superscript o denotes the optimal equilibrium allocation. Under fixed technology and constant prices, this equilibrium bundle can be represented as a function of Y^o and m_Y.

Substituting this representation into the revenue and cost functions implies that the equilibrium profit can be written as

$$\tilde{\pi} = (1 - \frac{1}{m_Y})pY^o.$$

In this equation, output Y^o can equivalently be represented as $\frac{F^{Yo}}{a_{YF}}$ according to the representation for fossil fuel-output ratio $a_{YF} = \frac{F^Y}{Y}$ in equilibrium. Thus, the above equation

can be written as

$$\tilde{\pi} = (1 - \frac{1}{m_Y}) \frac{pF^{Yo}}{a_{YF}}.$$

This representation highlights the role of fossil fuels in firm production and provides a direct link between fossil-fuel use and output. This link is relevant for Equation (12), where output contributes to the evolution of the CO_2 stock, which serves as the state variable in the DSGE model developed in Section 2.

The above discussion focuses on the manufacturing sector. The same analysis applies to the electricity sector.

A2.2. The Role of Initial Technologies in Technology Dynamics

This section analyzes technology by first establishing the asymptotic convergence of the R&D investment-to-technology ratios, then deriving technology dynamics as a function of existing technology and R&D investment, and finally characterizing equilibrium technology dynamics in a proposition.

R&D investment

R&D investment contributes to technological progress by providing inputs into knowledge production. Let R^Y denote the manufacturing sector's R&D investment and let R^E denote the electricity sector's R&D investment. Following Hall and Lerner (2010, Section 3), we emphasize the role of internal funds as an important source of R&D financing, particularly for R&D-intensive firms facing external financing constraints. For tractability, R&D investment is modeled as a fraction of sectoral profits: $R^j = \eta_R^j \pi^j, j \in \{Y, E\}$, where η_R^j captures the allocation of internal funds toward R&D activities. The benchmark specification sets $\eta_R^j = 1$. Because sectoral profits π^j depend on factor inputs, output, the market-structure parameter m_j, and sectoral technology (denoted by A^j), R&D investment is consequently related to the sector's technology level. Under the asymptotic relationship among the input-output coefficients characterized in Remark 1, the ratio R^j/A^j can converge to a stable value for a given market-structure parameter m_j. We denote this long-run ratio by a_{AR}^j and treat it as constant along the balanced-growth path.

Technology dynamics

Technology evolves with the accumulation of knowledge, while knowledge creation in the manufacturing and electricity sectors depends on existing technology and R&D investment. We

model knowledge production as Cobb-Douglas functions, $(A^Y)^{\nu_Y}(R^Y)^{1-\nu_Y}$ and $(A^E)^{\nu_E}(R^E)^{1-\nu_E}$, where A^Y and A^E denote current technological levels in the respective sectors. For tractability, human capital is omitted from the knowledge production function and is implicitly treated as a fixed input. Under this simplification, knowledge accumulation depends only on existing technology and R&D investment.

Since electricity is an input to manufacturing, technological improvements in the electricity sector can affect manufacturing productivity. In turn, advances in manufacturing technology can affect electricity-generation technology, for example through innovations in solar panel production that may accelerate the electricity sector's transition away from fossil fuels.

To capture these interactions, we draw on the real business cycle literature (e.g., Galí, 1999; Galí and Rabanal, 2004) and introduce cross-sectoral technology spillovers through the coupled stochastic processes specified below, together with the Cobb–Douglas knowledge production functions specified above:

$$\dot{A}^Y = (A^Y)^{\nu_Y}(R^Y)^{1-\nu_Y} + \gamma_Y^E A^E + \epsilon_A^Y$$

$$\dot{A}^E = (A^E)^{\nu_E}(R^E)^{1-\nu_E} + \gamma_E^Y A^Y + \epsilon_A^E$$

where ϵ_A^Y and ϵ_A^E denote mean-zero exogenous disturbance processes. Assume that the disturbance processes have zero conditional mean with respect to the information available when the shocks are realized. ϵ_A^Y and ϵ_A^E satisfy $\mathrm{E_t}[\epsilon_A^Y] = 0$, $\mathrm{E_t}[\epsilon_A^E] = 0$, $\mathrm{E_t}[\epsilon_A^Y \epsilon_A^E] = 0$, $\mathrm{E_t}[(\epsilon_A^Y)^2] = \sigma_A^Y$, and $\mathrm{E_t}[(\epsilon_A^E)^2] = \sigma_A^E$, with σ_A^Y and σ_A^E constant. The technology shocks are assumed to have zero conditional means and to be orthogonal to the information available when current R&D investment is determined. The coefficients $\gamma_Y^E \geq 0$ and $\gamma_E^Y \geq 0$ measure the strength of cross-sectoral knowledge spillovers between the manufacturing and electricity sectors and are treated as constant structural parameters.

Under this framework, long-run technological growth is jointly determined by sectoral R&D investment and cross-sectoral knowledge spillovers. However, because the specification exhibits diminishing returns to R&D capital, it does not generate the scale effects characteristic of first-generation endogenous growth models. Instead, it is consistent with semi-endogenous growth theory.

Equilibrium technology dynamics

Define $a_{11} = (a_{AR}^{Y})^{1-\nu_Y}$, $a_{12} = \gamma_Y^E$, $a_{21} = \gamma_E^Y$, and $a_{22} = (a_{AR}^{E})^{1-\nu_E}$, where $a_{AR}^{Y} = \frac{R^Y}{A^Y}$, $a_{AR}^{E} = \frac{R^E}{A^E}$. As discussed earlier in this appendix, a_{AR}^{Y} and a_{AR}^{E} converge to constants along the balanced-growth path. Let $\Lambda = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$. Under the maintained assumption that the R&D-to-technology ratios are constant along the balanced-growth path, the system becomes linear:

Proposition A2.1. Assume that the economy evolves along a balanced-growth path on which the ratios R^i/A^i remain constant for $i \in \{Y, E\}$, and suppose that Λ is diagonalizable. Let P satisfy $\Lambda P = PJ$, where $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$ and $J = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$. Then, the sectoral technologies evolve according to

$$A^Y = \left(p_{11}e^{\lambda_1 t}Z^Y(0) + p_{12}e^{\lambda_2 t}Z^E(0)\right) + \left(p_{11}\int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y ds + p_{12}\int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E ds\right)$$

$$A^E = \left(p_{21}e^{\lambda_1 t}Z^Y(0) + p_{22}e^{\lambda_2 t}Z^E(0)\right) + \left(p_{21}\int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y ds + p_{22}\int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E ds\right)$$

where

(i) $\begin{bmatrix} Z^Y \\ Z^E \end{bmatrix} \equiv P^{-1}\begin{bmatrix} A^Y \\ A^E \end{bmatrix}$ and $\begin{bmatrix} \epsilon_Z^Y \\ \epsilon_Z^E \end{bmatrix} \equiv P^{-1}\begin{bmatrix} \epsilon_A^Y \\ \epsilon_A^E \end{bmatrix}$;

(ii) $Z^i(0)$ are Z^i at time 0 for $i \in \{Y, E\}$.

(***See the proof at the end of this subsection.***)

In Proposition A2.1, the eigenvalues λ_1 and λ_2 characterize the dynamic modes of the coupled technology system and depend on sector-specific R&D productivity and cross-sectoral knowledge spillovers. The matrix P contains the corresponding eigenvectors and maps these system-wide dynamic modes into the observed sectoral technologies. Specifically, $A^Y = p_{11}Z^Y + p_{12}Z^E$ and $A^E = p_{21}Z^Y + p_{22}Z^E$.

Because $\mathrm{E_t}[\epsilon_A^Y]$=0 and $\mathrm{E_t}[\epsilon_A^E]$ =0, it follows that $\mathrm{E_t}[\epsilon_Z^Y]$=0 and $\mathrm{E_t}[\epsilon_Z^E]$=0. It follows that $\mathrm{E_0}[\int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_Z^Y ds]$ =0 and $\mathrm{E_0}[\int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_Z^E ds]$ =0. These zero-conditional-mean conditions imply that the conditional expected technology paths are determined by the initial technology conditions, transformed through P^{-1}, and the deterministic components of the dynamic system. As discussed earlier in this appendix, a stable ratio can arise between an economy's sectoral R&D investment and its sectoral technology level conditional on a given market-structure parameter. Consequently, cross-country differences in initial technology can persist, while differences in market structure can further generate heterogeneity in technology through their effects on R&D investment. Together, these mechanisms provide a theoretical basis for persistent cross-country heterogeneity in sectoral technology and are related to the mechanisms emphasized in the club-convergence literature (e.g., Durlauf and Johnson, 1995; Quah, 1996).

Sub-Appendix to Appendix A2.2: Proof of Proposition A2.1

To characterize the dynamics of sectoral technology, we begin with the technology accumulation equations presented in the text:

$$\dot{A}^Y = (A^Y)^{\nu_Y}(R^Y)^{1-\nu_Y} + \gamma_Y^E A^E + \epsilon_A^Y \quad \text{(A-6-1)}$$

$$\dot{A}^E = (A^E)^{\nu_E}(R^E)^{1-\nu_E} + \gamma_E^Y A^Y + \epsilon_A^E \quad \text{(A-6-2)}$$

where $\mathrm{E_t}[\epsilon_A^Y]$=0, $\mathrm{E_t}[(\epsilon_A^Y)^2]$=$\sigma_A^Y$, $\mathrm{E_t}[\epsilon_A^E]$=0, $\mathrm{E_t}[(\epsilon_A^E)^2]$=$\sigma_A^E$, and $\mathrm{E_t}[\epsilon_A^Y\epsilon_A^E]$=0.

Define $a_{AR}^{Y} = \frac{R^{Y}}{A^{Y}}$ and $a_{AR}^{E} = \frac{R^{E}}{A^{E}}$. We characterize the technology dynamics conditional on locally constant R&D-to-technology ratios. When these ratios are constant along the balanced-growth path, the resulting system is linear and admits the eigenvalue representation below. This characterization, however, does not necessarily extend to the global dynamics of the original nonlinear system.

We rewrite Equations (A-6-1) and (A-6-2) in matrix form as:

$$\begin{bmatrix} \dot{A}^{Y} \\ \dot{A}^{E} \end{bmatrix} = \begin{bmatrix} (a_{AR}^{Y})^{1-\nu_Y} & \gamma_{Y}^{E} \\ \gamma_{E}^{Y} & (a_{AR}^{E})^{1-\nu_E} \end{bmatrix} \begin{bmatrix} A^{Y} \\ A^{E} \end{bmatrix} + \begin{bmatrix} \epsilon_{A}^{Y} \\ \epsilon_{A}^{E} \end{bmatrix}. \tag{A-7}$$

Define $a_{11} = (a_{AR}^{Y})^{1-\nu_Y}$, $a_{12} = \gamma_{Y}^{E}$, $a_{21} = \gamma_{E}^{Y}$, and $a_{22} = (a_{AR}^{E})^{1-\nu_E}$. Let

$$\Lambda = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

so that

$$\begin{bmatrix} \dot{A}^{Y} \\ \dot{A}^{E} \end{bmatrix} = \Lambda \begin{bmatrix} A^{Y} \\ A^{E} \end{bmatrix} + \begin{bmatrix} \epsilon_{A}^{Y} \\ \epsilon_{A}^{E} \end{bmatrix} \tag{A-8}$$

Assume that Λ is diagonalizable. Then there exists a constant matrix P such that

$$\Lambda P = PJ, \text{ where } J = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \text{ and } P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}.$$

That is,

$$\Lambda = PJP^{-1}$$

Substituting into Equation (A-8) yields:

$$P^{-1} \begin{bmatrix} \dot{A}^{Y} \\ \dot{A}^{E} \end{bmatrix} = JP^{-1} \begin{bmatrix} A^{Y} \\ A^{E} \end{bmatrix} + P^{-1} \begin{bmatrix} \epsilon_{A}^{Y} \\ \epsilon_{A}^{E} \end{bmatrix} \tag{A-9}$$

Define

$$Z = P^{-1} \begin{bmatrix} A^{Y} \\ A^{E} \end{bmatrix} \equiv \begin{bmatrix} Z^{Y} \\ Z^{E} \end{bmatrix}$$

$$\tilde{\epsilon} = P^{-1} \begin{bmatrix} \epsilon_{A}^{Y} \\ \epsilon_{A}^{E} \end{bmatrix} \equiv \begin{bmatrix} \epsilon_{z}^{Y} \\ \epsilon_{z}^{E} \end{bmatrix}$$

where Z^{i}, $i \in \{Y, E\}$, are linear combinations of (A^{Y}, A^{E}), and ϵ_{z}^{i} are linear transformations

of $(\epsilon_A^Y, \epsilon_A^E)$, both induced by P^{-1}. Although the transformed disturbances are linear combinations of the original shocks, they remain mean-zero disturbances under the maintained assumptions.

Then Equation (A-9) becomes:

$$\dot{Z} = JZ + \tilde{\epsilon}$$

$$\Rightarrow \begin{bmatrix} \dot{Z}^Y \\ \dot{Z}^E \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} Z^Y \\ Z^E \end{bmatrix} + \begin{bmatrix} \epsilon_Z^Y \\ \epsilon_Z^E \end{bmatrix}$$

From this, we solve for $\dot{Z}^Y$:

$$\dot{Z}^Y = \lambda_1 Z^Y + \epsilon_Z^Y$$

$$\Rightarrow e^{-\lambda_1 t}\dot{Z}^Y = e^{-\lambda_1 t}\lambda_1 Z^Y + e^{-\lambda_1 t}\epsilon_Z^Y$$

$$\Rightarrow e^{-\lambda_1 t}\dot{Z}^Y - e^{-\lambda_1 t}\lambda_1 Z^Y = e^{-\lambda_1 t}\epsilon_Z^Y$$

$$\Rightarrow \int_{s=0}^{s=t} \frac{d(e^{-\lambda_1 s} Z^Y)}{ds} ds = \int_{s=0}^{s=t} e^{-\lambda_1 s}\epsilon_Z^Y(s) ds$$

$$\Rightarrow e^{-\lambda_1 t} Z^Y - Z^Y(0) = \int_{s=0}^{s=t} e^{-\lambda_1 s}\epsilon_Z^Y(s) ds$$

Thus,

$$Z^Y = e^{\lambda_1 t} Z^Y(0) + \int_{s=0}^{s=t} e^{\lambda_1 (t-s)}\epsilon_Z^Y(s) ds \qquad \text{(A-10)}$$

Similarly,

$$Z^E = e^{\lambda_2 t} Z^E(0) + \int_{s=0}^{s=t} e^{\lambda_2 (t-s)}\epsilon_Z^E(s) ds \qquad \text{(A-11)}$$

Finally, since $P^{-1}\begin{bmatrix} A^Y \\ A^E \end{bmatrix} \equiv \begin{bmatrix} Z^Y \\ Z^E \end{bmatrix}$ (Equation (A-9)), we obtain from Equations (A-10) and (A-11):

$$\begin{bmatrix} A^Y \\ A^E \end{bmatrix} = P \begin{bmatrix} e^{\lambda_1 t} Z^Y(0) + \int_{s=0}^{s=t} e^{\lambda_1 (t-s)}\epsilon_Z^Y(s) ds \\ e^{\lambda_2 t} Z^E(0) + \int_{s=0}^{s=t} e^{\lambda_2 (t-s)}\epsilon_Z^E(s) ds \end{bmatrix}$$

$$= \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} e^{\lambda_1 t} Z^Y(0) + \int_{s=0}^{s=t} e^{\lambda_1 (t-s)}\epsilon_Z^Y(s) ds \\ e^{\lambda_2 t} Z^E(0) + \int_{s=0}^{s=t} e^{\lambda_2 (t-s)}\epsilon_Z^E(s) ds \end{bmatrix}$$

$$= \begin{bmatrix} p_{11}(e^{\lambda_1 t}Z^Y(0) + \int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y(s)ds) + p_{12}(e^{\lambda_2 t}Z^E(0) + \int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E(s)ds) \\ p_{21}(e^{\lambda_1 t}Z^Y(0) + \int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y(s)ds) + p_{22}(e^{\lambda_2 t}Z^E(0) + \int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E(s)ds) \end{bmatrix}$$

$$= \begin{bmatrix} \left(p_{11}e^{\lambda_1 t}Z^Y(0) + p_{12}e^{\lambda_2 t}Z^E(0)\right) + \left(p_{11}\int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y(s)ds + p_{12}\int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E(s)ds\right) \\ \left(p_{21}e^{\lambda_1 t}Z^Y(0) + p_{22}e^{\lambda_2 t}Z^E(0)\right) + \left(p_{21}\int_{s=0}^{s=t} e^{\lambda_1(t-s)}\epsilon_z^Y(s)ds + p_{22}\int_{s=0}^{s=t} e^{\lambda_2(t-s)}\epsilon_z^E(s)ds\right) \end{bmatrix}$$

Q.E.D.

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Appendix B: Data, Empirical Results, and Robustness Analyses

Appendix B1: Data and Variable Definitions

The Bayesian system in this study consists of the macroeconomic block, described by Equations (10), (11'), and (12), and the sectoral block, which comprises Equations (11-1) and (13). Following the normalization and transformation practices commonly used in DSGE estimation (e.g., An and Schorfheide, 2007), the observed series are rescaled for numerical implementation and estimation. This rescaling is applied only to the numerical implementation of the estimation procedure and does not alter the underlying structural equations or the economic interpretation of the structural parameters.

The macroeconomic block includes control variables (C_t and Y_t) and a state variable (D_t). We represent C_t and Y_t using data on household consumption per capita and gross domestic product per capita (constant 2015 USD), respectively, from the World Bank (2025), World Development Indicators (WDI).[2] The cumulative sum of carbon dioxide emissions from the manufacturing and electricity sectors since 1990 is used as the proxy for unscaled D_t.[3] The source of carbon dioxide data is the EPA (2025, data).[4] It is important to note that the variables C_t, Y_t, and D_t used in our analysis are appropriately standardized by subtracting their respective means and dividing by their respective standard deviations.

The macroeconomic block is linked to the sectoral block through the value of Ω_1 inferred from it. Within the sectoral block, prices (p and e), the emission absorption factor (τ_{n0}), and input–output coefficients and intensity measures (a_{YE}, a_{YF}, b_{EF}, and v_K) are used to estimate

[2] The full name of per capita household consumption from the World Development Indicators is households and non-profit institutions serving households (NPISHs) final consumption expenditure per capita (constant 2015 US$).

[3] We accumulate carbon dioxide from Year 1990.

[4] The data on the carbon emissions due to manufacturing is from EPA's (2025) Table frifj2-1, and the data on that due to energy use is from EPA's (2025) Table frifj1-1.

revenue-to-cost ratios (m_E and m_Y), the fossil-fuel price (f), and effective capital-replenishment coefficients (κ_K and κ_E).

The commodity price p and electricity price e are represented by the U.S. consumer price index from WDI (2025) and the annual U.S. industrial electricity price from EIA (2025b), respectively. The consumer price index is used as an empirical proxy for p because no directly corresponding economy-wide price series is available. Price e standardized in the same manner as C_t, Y_t, and D_t by dividing the demeaned value by its standard deviation. In contrast, although p is reasonably centered around 100, we normalize it to approximately 1 by scaling it down by a factor of 100. This normalization is applied consistently wherever p enters the equilibrium mappings.

For τ_{n0}, which represents the ratio of carbon dioxide absorbed by natural systems to the total carbon emissions from the electricity and manufacturing sectors, no further scaling is applied to τ_{n0}, because it is already expressed as a ratio bounded between 0 and 1. Data for emissions and carbon sink are obtained from the EPA (2025, data).[5]

The coefficients a_{YF} and b_{EF} are computed as $a_{YF} = \frac{F^Y}{Y}$ and $b_{EF} = \frac{F^E}{E^Y}$, respectively, using coal consumption (F) from EIA (2025d), real GDP (Y) from WDI (2025), and electricity consumption (E^Y) from EIA (2025c). Because a consistent annual series for manufacturing electricity consumption is unavailable over the full sample period, manufacturing electricity input is proxied by industrial-sector electricity consumption.

v_K is computed as the ratio of investment to real manufacturing value added, as discussed in Equation (8). Because gross investment is not directly available at the required sectoral

[5] In EPA (2025), Table frifj1-1 of Chapter 3 is for the carbon emissions sourced from the electricity sector, Table frifj2-1 of Chapter 4 for the carbon emissions sourced from the manufacturing industry, and Table 6-2 of Chapter 6 for carbon sinks of the world.

frequency, the change in the real capital stock is used as an empirical proxy for investment, with the underlying capital stock data obtained from the BEA (2025).[6]

Table B1 provides summary statistics for the variables used in the analysis. These variables are based on annual data, and the analysis covers the period from 1990 to 2024.

[6] We compute the economy's total physical capital to be the sum of capital in electricity and manufacturing sectors.

Table B1. Summary statistics for annual data,1990−2024

Variable	Obs	Mean	Standard Deviation
Unstandardized Y	34	51.52155	7.640364
Unstandardized C	35	32.19377	8.665413
Unstandardized D	33	94923.54	54781.06
Unscaled e	31	5.771613	1.021529
Unscaled p	34	93.5212	21.3442
Unstandardized E	33	3407.801	109.2113
Unstandardized F of the manufacturing sector	33	78202	22743.05
Unstandardized F of the electricity sector	33	839874.8	173055.3
Unstandardized K of the manufacturing sector	34	157548.7	27571.8
Unstandardized K of the electricity sector	34	1594777	376383.7
τ_{n0}	33	0.1809	0.014638

Notes:

(1) Unstandardized C and Y represent household consumption per capita and gross domestic product per capita (constant 2015 USD) from World Bank (2025), World Development Indicators. Unscaled p is the U.S. consumer price index from the WDI. Unstandardized K is the fixed-cost net capital stock of private nonresidential fixed assets from the BEA (2025). Unscaled e is the annual U.S. electricity price from the EIA (2025b). Unstandardized E is industrial-sector electricity consumption (EIA, 2025c), and unstandardized F is coal consumption (EIA, 2025d). The CO_2 stock (D) is computed as the cumulative sum of carbon dioxide emissions from the manufacturing and electricity sectors since 1990, serving as the proxy for unscaled D. CO_2 data are sourced from the EPA (2025).

(2) We compute τ_{n0} to be the ratios of carbon dioxide absorbed by nature to the total carbon emissions of the electricity and manufacturing sectors. τ_{n0} is not scaled, and the data source of emissions and carbon sink is the EPA (2025).

(3) Units: Unstandardized Y: thousand USD per capita (constant 2015); Unstandardized C: thousand USD per capita (constant 2015); Unstandardized D: million metric tons CO_2 (cumulative since 1990; end year=2024); Unscaled e: cents/kWh; Unstandardized E: trillion Btu; Unstandardized F: thousand short tons; Unstandardized K: million USD (constant prices).

Data sources

An, S., & Schorfheide, F. (2007). Bayesian analysis of DSGE models. *Econometric Reviews* 26(2-4), 113-172.

BEA (2025) Fixed-(Current-)Cost Net Capital Stock of Private Nonresidential Fixed Assets. U.S. Bureau of Economic Analysis. Available at: https://apps.bea.gov/iTable/?ReqID=10&step=2&_gl=1*k8z4v1*_ga*Mjc1MDg1Mjg4LjE3NDQ4MTA5MTg.*_ga_J4698JNNFT*MTc0NTMzNjQzMy4yLjEuMTc0NTMzNzMzNi41Ni4wLjA

EIA (2025a) Carbon Dioxide Emissions Coefficients. Available at: https://www.eia.gov/environment/emissions/CO_2_vol_mass.php

EIA (2025b) Average Price of Electricity by State by Provider – U.S. Energy Information Administration. Data.gov. Available at: https://catalog.data.gov/dataset/annual-average-electricity-price-by-state-by-provider-from-1990-latest-year-available/resource/126c8c5d-f9a5-457a-9849-d48aa596df5b

EIA (2025c) Table 2.1a: Energy Consumption: Residential, Commercial, and Industrial Sectors. Total Energy-Annual Energy Review. U.S. Energy Information Administration. Available at: https://www.eia.gov/totalenergy/data/annual/

EIA (2025d) Table 6.2 Coal Consumption by Sector. Annual Energy Review. U.S. Energy Information Administration. Available at: https://www.eia.gov/totalenergy/data/annual/

EPA (2025) Inventory of U.S. Greenhouse Gas Emissions and Sinks. U.S. Environmental Protection Agency. Available at: https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks

World Bank (2025) World Development Indicators 2025. World Bank. Available at: https://data.worldbank.org/indicator

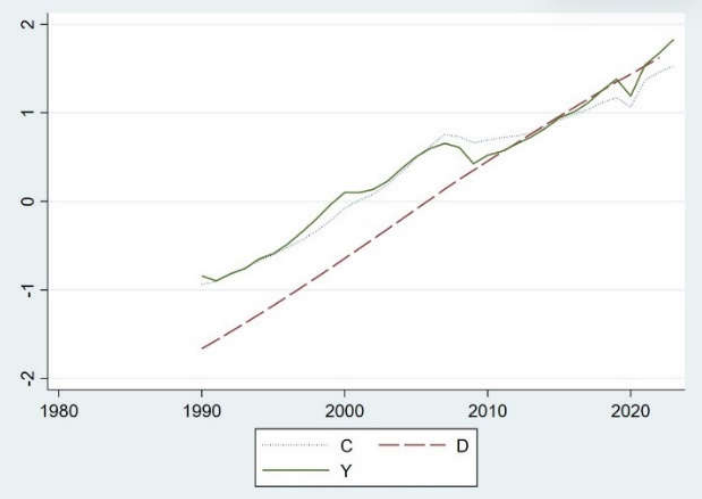
Appendix B2: Stylized Analysis of Carbon Dynamics

Carbon dynamics (as expressed in Equation (12)) involve three variables, C, Y, and D, which are standardized prior to estimation to improve numerical stability in the Bayesian solution algorithm and to ensure comparability across variables measured in different units. The standardized series plotted in Figure B1 exhibit broadly similar upward trends. Given the carbon-stock dynamics in Equation (12), positive output and the associated emissions can contribute to continued carbon accumulation when emissions exceed natural absorption. Thus, declining current emissions do not necessarily imply a decline in the accumulated CO_2 stock.

In Equation (12-1), the conversion rate Ω_3 of manufacturing output into CO_2 emissions is influenced by its emissions factor. In the United States, the coal emissions factor in the electricity sector has remained highly stable, changing only slightly from 95.11 MMT CO_2 per quadrillion Btu in 1973 to 95.81 MMT in 2022;[7] similarly, the industrial coal emissions coefficient increased modestly from 94.12 MMT to 94.80 MMT over the same period (EIA, 2025a, Carbon Dioxide Emissions Coefficients Table). Moreover, the ratio of the fossil-fuel input-output coefficient in the electricity sector to that in the manufacturing sector has a mean of 0.91 and a standard deviation of 0.02 between 1990 and 2024. Overall, these results suggest remarkably stable pollution emission and energy conversion efficiencies over time. The observed persistence may be associated with potential path dependence in U.S. energy technology adoption (Makholm, 2025).

[7] The number 94.12 is calculated as the average of the coefficient for industrial coke plants in 1973 (93.61) and the coefficient for other industrial uses in the same year (94.64). The value 94.80 is determined by averaging the coefficients for industrial coke plants (93.90) and other industrial uses in the year 2022 (95.70).

Figure B1. Paths of standardized Y, C and D



Note: horizontal axis= year.

References for Appendix B2

Makholm, J. D. (2025) What doesn't work: Path dependence in U.S. electricity supply in 2025. *Climate and Energy* 41(8), 20–28.

Appendix B3: Bayesian Estimation Results and Economic Implications

B3.1. Macroeconomic block: parameter estimates and carbon capture requirements

This subsection presents the estimation results of the macroeconomic block, derives the implied coefficients associated with CCS and vegetation restoration, and discusses their economic implications for long-run sustainability. Panel A of Table B2 lists the parameter estimates. The small MCSEs indicate accurate numerical approximation of the posterior distributions. Posterior diagnostic plots in **Appendix C1.1**. Limited persistence in the MCMC draws is indicated by the rapid decay of the autocorrelation values for Ω_0, Ω_1, and Ω_2 to approximately zero within the first 10 lags of the 40-lag window, although Ω_3 and Ω_4 remain slightly above zero by the end of the window. Together with the trace plots, posterior densities, acceptance rates, and small MCSEs, these diagnostics indicate satisfactory numerical mixing and stability of the baseline posterior simulations.

The estimated value of Ω_0 is 0.5710. As Equation (10) is a reduced-form representation of the Euler equation with standardized consumption, Ω_0 is an equilibrium parameter governing consumption dynamics rather than a structural discount-related parameter. A value of Ω_0 below unity implies mean-reverting consumption dynamics.

Larger Ω_2 (0.4456) than Ω_1 (0.3980) suggests a stronger response of consumption to changes in the accumulated CO_2 stock than to comparable changes in output. Larger Ω_3 (0.2856) than Ω_4 (0.0981) implies faster CO_2 accumulation than natural removal. Thus, long-run sustainability requires either lower output growth or stronger mitigation. Taken together, these estimates from Equations (11) and (12) imply that sustained output growth without sufficient mitigation may offset some consumption gains through CO_2 accumulation.

We derive τ_s and τ_{nx} from Ω_3 and Ω_4 and report them in Panel B of Table B2, with details provided in **Appendix C3**. The estimated τ_s =0.7144 implies that CCS would need to capture approximately 71.44% of carbon emissions under the baseline trajectory. Given the costs associated with large-scale CCS deployment (Baylin-Stern and Berghout, 2021), relying exclusively on CCS would entail substantial economic costs.

The estimated $\tau_{nx} = -0.0828$ implies a negative contribution of the anthropogenic component to natural carbon absorption. This effect amounts to 45.77% of the baseline absorption coefficient τ_{n0} reported in Table B1 **(Appendix B1)**, suggesting substantial potential for vegetation restoration to recover natural carbon sequestration and reduce reliance on CCS and the associated long-run sustainability costs.

B3.2. Sectoral block: market structure and welfare implications

In this subsection, we examine the sector-level parameters of the sectoral block. Table B3 reports the estimated values: f =0.2855, m_E =1.6211, m_Y =1.4527, κ_K =0.2279, and κ_E=0.2470. Diagnostic plots are provided in **Appendix C1.2**. Autocorrelation plots in the appendix suggest that the autocorrelations of m_Y and κ_E decay to approximately zero within 10 lags, while those of m_E, f, and κ_K decay within approximately 20 lags. The trace plots, posterior densities, acceptance rates, and small MCSEs provide complementary evidence that the baseline posterior simulations mix satisfactorily and yield numerically stable estimates.

The positive fossil fuel price, f, suggests persistent demand for fossil fuels. This implies that, even if the U.S. economy achieves its sustainability objective within the DSGE framework, fossil fuels remain a necessary input in production.

The revenue-to-cost ratios for the electricity and manufacturing sectors, m_E and m_Y, respectively, exceed unity, indicating lower market competitiveness and higher profits available for R&D activities. Because m_E and m_Y are components of Ω_1 (Equation (11-1)), which in turn affects consumption through manufacturing consumption, higher values of these ratios are

therefore associated with lower modeled consumption through their effect on C (Equation (11')).

Finally, a higher κ_E (=0.2470) relative to κ_K (=0.2279) indicates that capital in the electricity sector has a higher effective capital-replenishment coefficient than capital in manufacturing. Since electricity-sector capital enters manufacturing's investment requirements through electricity intensity, this indicates a higher effective capital-replenishment requirement per unit of electricity-sector capital involved in production.

Table B2. Macroeconomic block parameters and estimates for CCS and vegetation restoration

A. Macroeconomic block			
Coefficient	Mean	SD	MCSE
Ω_0	0.5710	0.0544	0.0011
Ω_1	0.3980	0.0564	0.0011
Ω_2	0.4456	0.0521	0.0010
Ω_3	0.2856	0.0531	0.0016
Ω_4	0.0981	0.0675	0.0028
B. Estimates of τ_{nx} and τ_s			
τ_{nx}	-0.0828		
τ_s	0.7144		

Notes:

(i) Parameter Ω_0 has a Beta(40,55) prior with the expected value of 40/(40+55)=0.42, parameter Ω_1 has a Beta(30,45) prior with the expected value of 30/(30+45)=0.4, parameter Ω_2 has a Beta(40,50) prior with the expected value of 40/(40+50)=0.44, parameter Ω_3 has a Beta(20,50) prior with the expected value of 20/(20+50)=0.29, and parameter Ω_4 has a N(0.1,0.005) prior with a mean of 0.1 and a standard deviation of 0.005. The adaptation tolerance is set to 0.209. The initial value of the model is set to β=0.45.

(ii) The Bayesian DSGE model is estimated using 34 observations. The MCMC procedure is based on 10,000 draws, with the first 1,000 discarded as burn-in and the remaining 9,000 used to approximate the posterior distribution. To reduce serial correlation, every 10th draw is retained from the chain. The average efficiency of the sampler is 0.1647. No significant autocorrelation is detected beyond 1,000 lags. The acceptance rate is 0.3428. Diagnostic plots in **Appendix C1.1** indicate satisfactory convergence of the Markov chains. The identification results reported later in the paper further suggest that the model is reasonably well identified given the available data.

(iii) $\tau_{nx} = \Omega_4 - \tau_{n0}$ and $\tau_s = 1 - \Omega_3$.

Table B3. Sectoral block estimates conditional on Ω_1

	Mean	SD	MCSE
m_E	1.6211	0.4783	0.0143
m_Y	1.4527	0.2714	0.0050
f	0.2855	0.0377	0.0008
κ_K	0.2279	0.0321	0.0008
κ_E	0.2470	0.0338	0.0007

Notes:

(i) Parameter m_E has a N(1.3,0.4) prior with the expected value of 1.3, parameters m_Y has a N(1.4,0.1) prior with the expected value of 1.4, and parameters f, κ_K and κ_E have the same Beta(40,100) prior with the expected value of 40/(40+100)=0.29. The prior distribution of the commodity price in Ω_1 follows a normal distribution, N(11.75, 0.01), with a mean of 11.75 and a standard deviation of 0.1. Similarly, the prior distribution of v_K follows a normal distribution, N(0.35, 0.01), with a mean of 0.35 and a standard deviation of 0.1. The initial values of κ_K, κ_E, and m_E are set to 0.3, 0.19, and 1.3, respectively, while the initial value of v_K is set to 0.45.

(ii) The sectoral block is based on multiple data series with sample sizes ranging from 18 to 33 observations, with an average of 26 observations per series. The MCMC procedure uses 10,000 draws, with the first 5,000 discarded as burn-in. To reduce autocorrelation, every 10th draw is retained for posterior approximation.

(iii) The adaptation tolerance is set to 0.23. No significant autocorrelation is detected beyond 2,000 lags. The acceptance rate is 0.2997, and the average efficiency is 0.2399. Diagnostic plots in **Appendix C1.2**, together with the identification structure of the model, suggest that the posterior distributions are stable and reasonably well identified, given the available sample information.

Appendix B4: MCMC Simulation-Size Robustness and Numerical Stability

In this section, we conduct a robustness check with respect to the MCMC simulation size by comparing posterior estimates obtained under baseline and enlarged numbers of draws. In Table B4, we compare the estimates of Ω_0, Ω_1, Ω_2, Ω_3, and Ω_4 for alternative MCMC sample sizes ranging from 15,000 to 40,000 with those reported in Table B2, which uses an MCMC sample size of 10,000. Specifically, the alternative cases use MCMC sample sizes of 15,000, 20,000, 25,000, 30,000, 35,000, and 40,000. The parameter ratios remain consistently close to 1, despite the increased MCMC sample size. This stability provides evidence that the baseline posterior estimates are not materially affected by the number of MCMC draws.

In a similar fashion to Table B4, Table B5 compares the estimates of m_E, m_Y, f, κ_K, and κ_E with larger MCMC sample sizes, exceeding 10,000, against those in Table B3 with an MCMC sample size of 10,000. Again, parameter ratios close to unity indicate that the posterior estimates of the sectoral block are not materially affected by the number of MCMC draws.

Table B4. Posterior stability under alternative MCMC simulation sizes (macroeconomic block)

	Ω_{0i}/Ω_{00}	Ω_{1i}/Ω_{10}	Ω_{2i}/Ω_{20}	Ω_{3i}/Ω_{30}	Ω_{4i}/Ω_{40}
MCMC=15,000	0.999975	1.000341	0.999159	0.999201	0.999939
MCMC=20,000	0.999883	1.001949	0.999818	1.004277	0.995905
MCMC=25,000	1.002614	1.000124	1.000573	1.007349	1.013449
MCMC=30,000	1.003053	0.999792	1.0008	1.007739	1.027066
MCMC=35,000	1.002184	0.999975	1.000537	1.006347	1.017069
MCMC=40,000	1.001293	1.001003	1.000189	1.006061	1.00929

Note: This table compares the parameter estimates from cases with larger MCMC sample sizes with those from the baseline case with 10,000 draws. The subscript i denotes an alternative MCMC simulation size and the subscript 0 denotes the baseline simulation with 10,000 draws.

Table B5. Posterior stability under alternative MCMC simulation sizes (sectoral block)

	m_{Ei}/m_{E0}	m_{Yi}/m_{Y0}	f_i/f_0	κ_{Ki}/κ_{K0}	κ_{Ei}/κ_{E0}
MCMC=15000	0.996339	0.998111	0.998856	1.000314	0.996556
MCMC=20000	0.994915	0.998934	0.997939	1.002147	0.995121
MCMC=25000	0.991012	0.999901	0.999178	1.002058	0.994264
MCMC=30000	0.987933	0.998952	1.000143	1.002226	0.994127
MCMC=35000	0.990255	0.998842	0.998605	1.003287	0.992343
MCMC=40000	0.990009	0.99782	0.998458	1.003266	0.991754

Note: See note for Table B4.

Appendix B5: Extended-Sample Analysis

The DSGE analysis in the main text is conducted using data for the period 1990-2024. This sample restriction arises from the unavailability of data on carbon stock and electricity prices prior to 1990.

Nevertheless, the missing carbon stock series can be reconstructed back to 1980 for the re-estimation of the macroeconomic block. We set the disturbance term to zero and use the observed value of carbon stock in 1990 as the boundary condition. The pre-1990 values of D are then recursively reconstructed backward using Equation (12), the estimated values of Ω_3 and Ω_4, and the observed series of Y. Similarly, the missing electricity price series prior to 1990 can be reconstructed for the re-estimation of the sectoral block. Based on Equation (11-1), the reconstruction of the electricity price e uses its observed value in 1990 as a boundary condition, along with the estimated values of m_E, m_Y, and f reported in Table B3, the observed series of p, b_{EF}, a_{YE}, a_{YF}, and v_K, and the plug-in parameter Ω_1 obtained from the macroeconomic block.

By reconstructing the pre-1990 carbon stock and electricity price series, the available sample period for each block is extended from 1990–2024 to 1980–2024, corresponding to a nominal sample length of 45 years. The effective sample sizes may differ across series because of lagged variables and other data requirements. This exercise is related in spirit to the sample-expansion experiments of Koop et al. (2013) and Chadha and Shibayama (2018). However, because the pre-1990 observations are model-reconstructed rather than independently observed, this exercise is interpreted as a sensitivity analysis to sample extension rather than as evidence from an independent historical sample.

Table B6 reports the re-estimation results for the macroeconomic block. The estimates of Ω_0, Ω_1, Ω_2, Ω_3, and Ω_4 remain broadly consistent with their counterparts in Table B2 in

terms of magnitude, sign, and relative relationships. The sectoral estimates reported in Table B7 show a similar qualitative pattern relative to Table B3.

We further examine the rank conditions of the two blocks using the SVD of the Jacobian matrices of the moment conditions with respect to the parameters. For the macroeconomic block, the singular values are 2.2474, 2.1555, 1.5871, 0.0099, and 0.0099. For the sectoral block, the corresponding singular values are 2.9125, 1.5697, 1.5111, 0.5317, and 0.2448. Each Jacobian has five columns corresponding to the five estimated parameters. Since all singular values are positive, the Jacobian matrices have full column rank, indicating local identification of both blocks.

The ratios of the largest to the smallest singular values are 227.0101(=2.2474/0.0099) and 11.8975(=2.9125/0.2448) for the macroeconomic and sectoral blocks, respectively. The substantially larger condition number for the macroeconomic block indicates weaker identification strength and greater numerical sensitivity relative to the sectoral block.

References for Appendix B5

Chadha, J. S. and K. Shibayama (2018) Bayesian Estimation of DSGE models: Identification using a diagnostic indicator. *Journal of Economic Dynamics & Control* 95, 172-186.

Koop, G., M. H. Pesaran, and R. P. Smith (2013) On identification of Bayesian DSGE models. *Journal of Business and Economic Statistics* 31, 300-314.

Table B6. Macroeconomic block parameters and estimates for the extended sample

	Mean	SD	MCSE
Ω_0	0.6699	0.0526	0.0018
Ω_1	0.3983	0.0561	0.0009
Ω_2	0.4440	0.0514	0.0008
Ω_3	0.2870	0.0542	0.0023
Ω_4	0.1002	0.0697	0.0079

Notes:

(i) Parameter Ω_0 has a Beta(40,55) prior with the expected value of 40/(40+55)=0.42, parameter Ω_1 has a Beta(30,45) prior with the expected value of 30/(30+45)=0.4, parameter Ω_2 has a Beta(40,50) prior with the expected value of 40/(40+50)=0.44, parameter Ω_3 has a Beta(20,50) prior with the expected value of 20/(20+50)=0.29, and parameter Ω_4 has a N(0.1,0.005) prior with a mean of 0.1 and a standard deviation of 0.005. The adaptation tolerance is set to 0.209. The initial value of the model is set to β=0.45.

(ii) The Bayesian DSGE model is estimated using 45 observations. The MCMC procedure is based on 10,000 draws. Every 20th MCMC draw is retained for posterior analysis to reduce the number of stored draws and potentially reduce autocorrelation among retained draws. The first 5,000 MCMC iterations are discarded to reduce the influence of the initial parameter values on posterior inference. The average efficiency of the sampler is 0.1552. No significant autocorrelation is detected beyond 1,000 lags. The acceptance rate is 0.3815.

Table B7. Sectoral block estimates conditional on Ω_1 for the extended sample

	Mean	SD	MCSE
m_E	1.1020	0.1004	0.0011
m_Y	1.5492	0.0842	0.0011
f	0.2841	0.1581	0.0017
κ_K	0.2835	0.1597	0.0017
κ_E	0.2862	0.1599	0.0011

Notes:

(i) Parameter m_E has a N(1.1,0.01) prior with the expected value of 1.1, parameters m_Y has a N(1.01,0.01) prior with the expected value of 1.01, and parameters f, κ_K and κ_E have the same Beta(2,5) prior with the expected value of 2/(2+5)=0.29. The prior distribution of the commodity price in Ω_1 follows a beta distribution, Beta(10,10), with a mean of 10/(10+10)=0.5. The prior distribution of v_K follows a beta distribution, Beta(50,50), with a mean of 50/(50+50)=0.5. The initial values of κ_K, κ_E, and m_E are set to 0.3, 0.3, and 1.1, respectively.

(ii) The sectoral block is based on multiple data series with sample sizes ranging from 41 to 43 observations, with an average of 42 observations per series. The MCMC procedure uses 50,000 draws. Every 10th MCMC draw is retained for posterior analysis to reduce the number of stored draws and potentially reduce autocorrelation among retained draws. The first 10,000 MCMC iterations are discarded to reduce the influence of the initial parameter values on posterior inference.

(iii) The adaptation tolerance is set to 0.234. No significant autocorrelation is detected beyond 2,000 lags. The acceptance rate is 0.264, and the average efficiency is 0.2187. The extended-sample exercise is intended as a sensitivity analysis to the inclusion of reconstructed pre-sample observations rather than as an independent validation of the baseline estimates.

Appendix C: Supplementary Estimation Diagnostics and Model Specification

Appendix C1: Bayesian Diagnostics

C1.1. Posterior Diagnostic Plots for the Macroeconomic Block

Note: Each figure in this appendix includes a trace plot, a histogram, an autocorrelation plot, and a posterior density plot. A trace plot illustrates how a parameter's estimated values evolve over the runtime of the chain. A histogram visually presents the distribution of the estimated parameter, while an autocorrelation plot depicts the correlation between an estimated parameter value and its preceding estimate. The posterior density plot provides insight into the distribution of estimated parameter values within the chain.

Figure C1. Posterior diagnostic plot for Ω_0

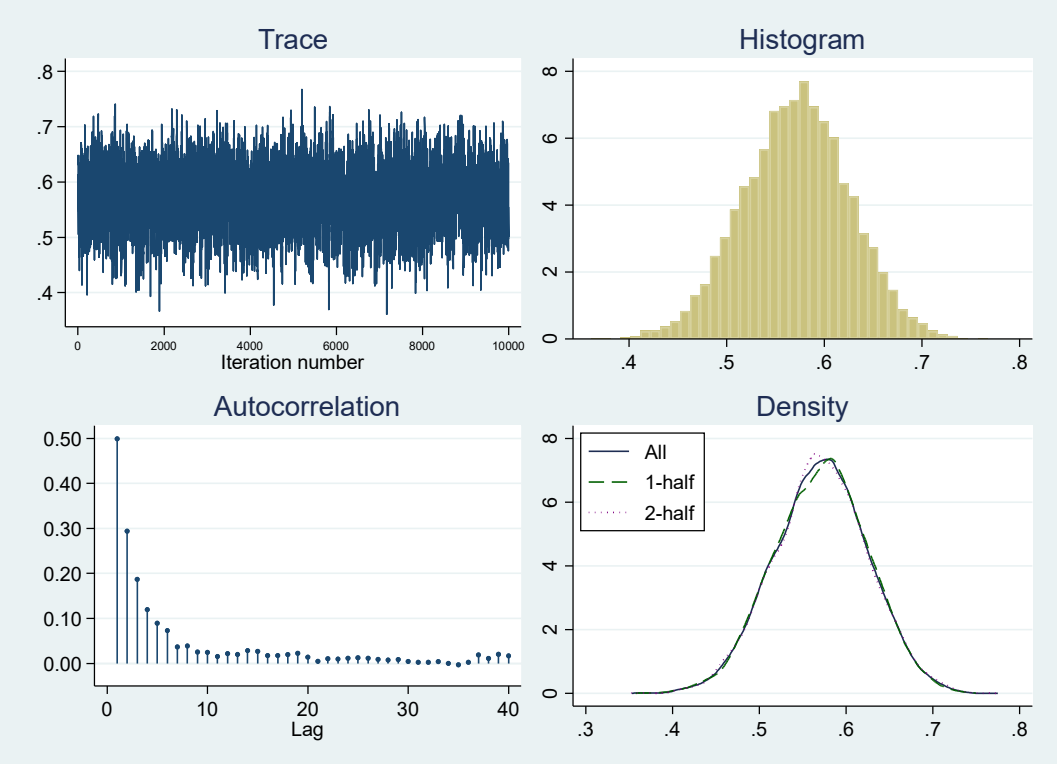


Figure C2. Posterior diagnostic plot for Ω_1

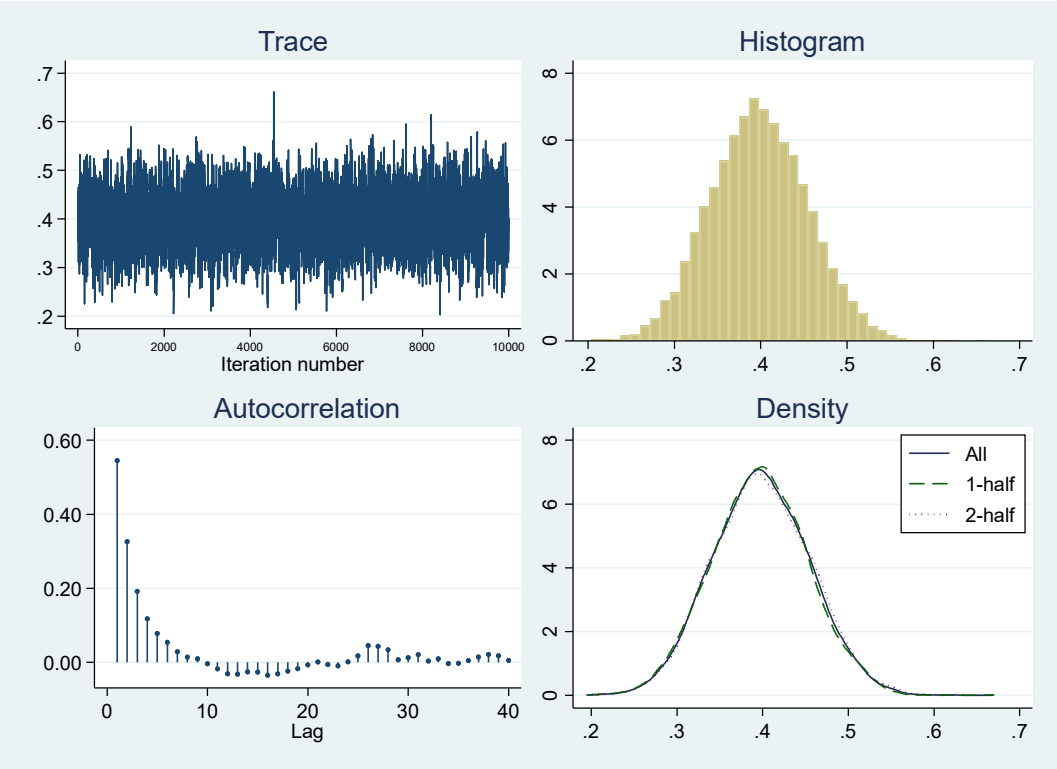


Figure C3. Posterior diagnostic plot for Ω_2

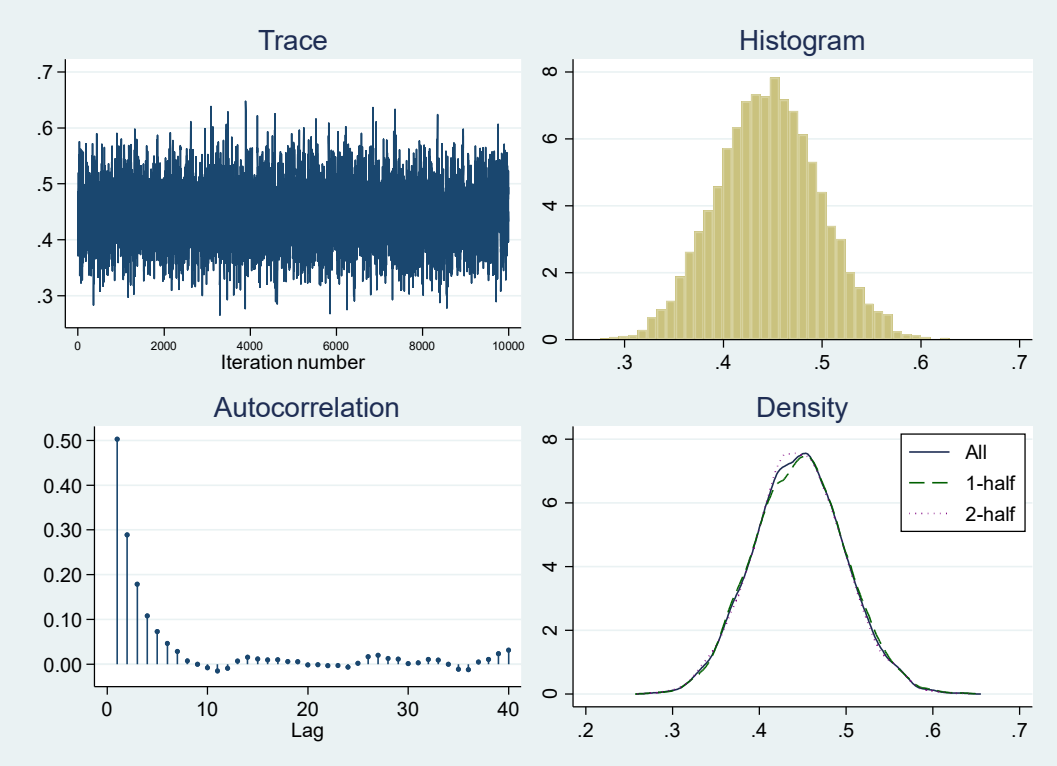


Figure C4. Posterior diagnostic plot for Ω_3

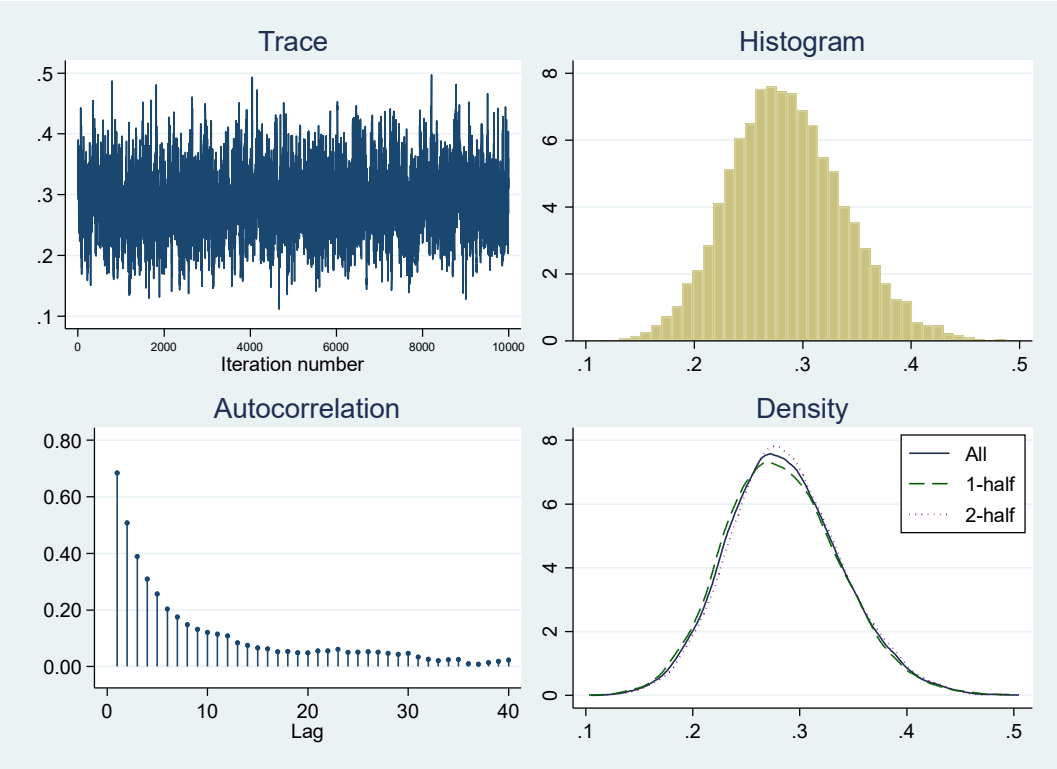
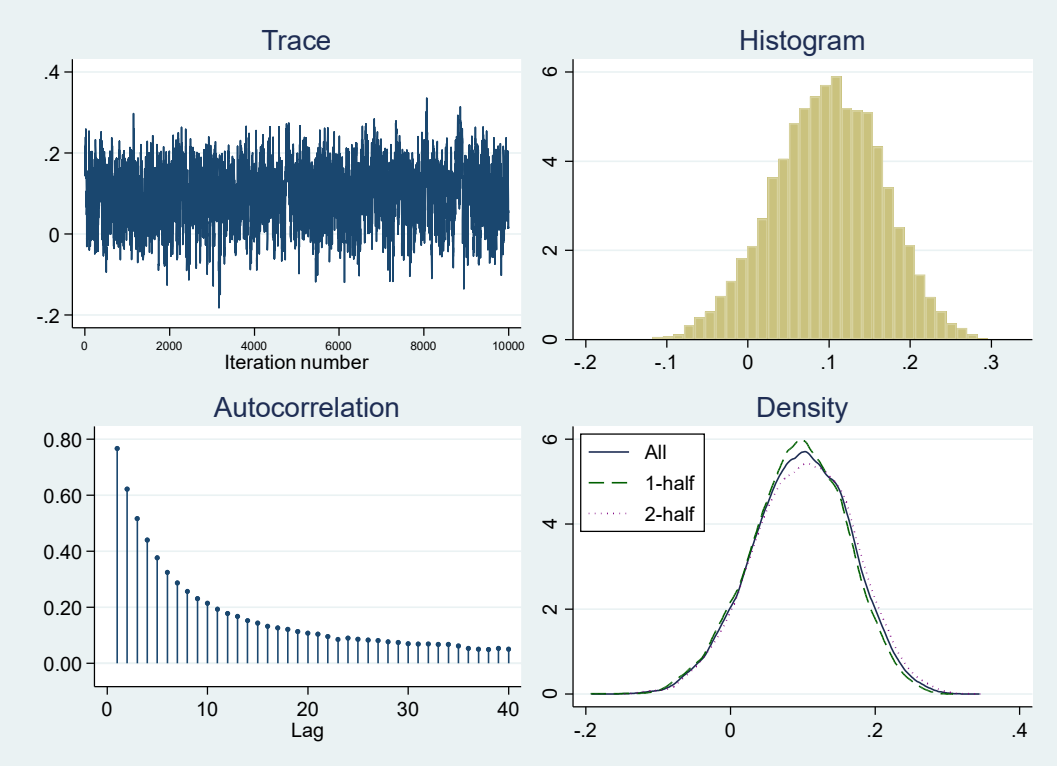


Figure C5. Posterior diagnostic plot for Ω_4



C1.2. Posterior Diagnostic Plots for the Sectoral Block

Note: For notes to this subsection, please refer to **Appendix C1.1**.

Figure C6. Posterior diagnostic plot for m_E

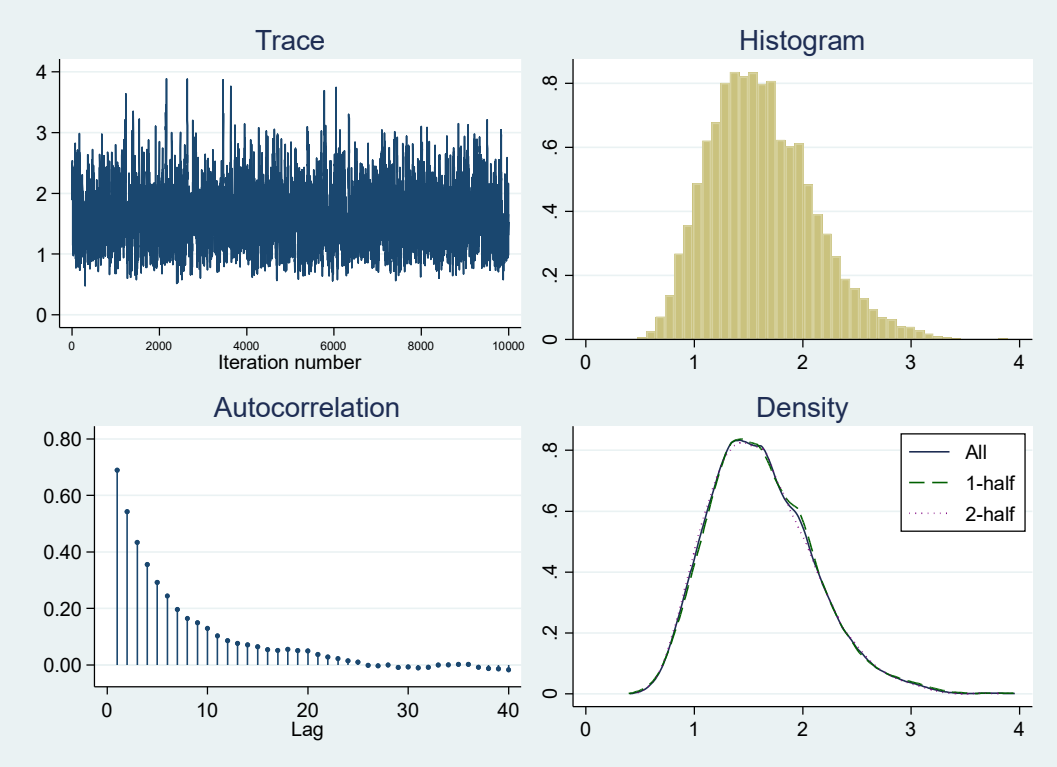


Figure C7. Posterior diagnostic plot for m_Y

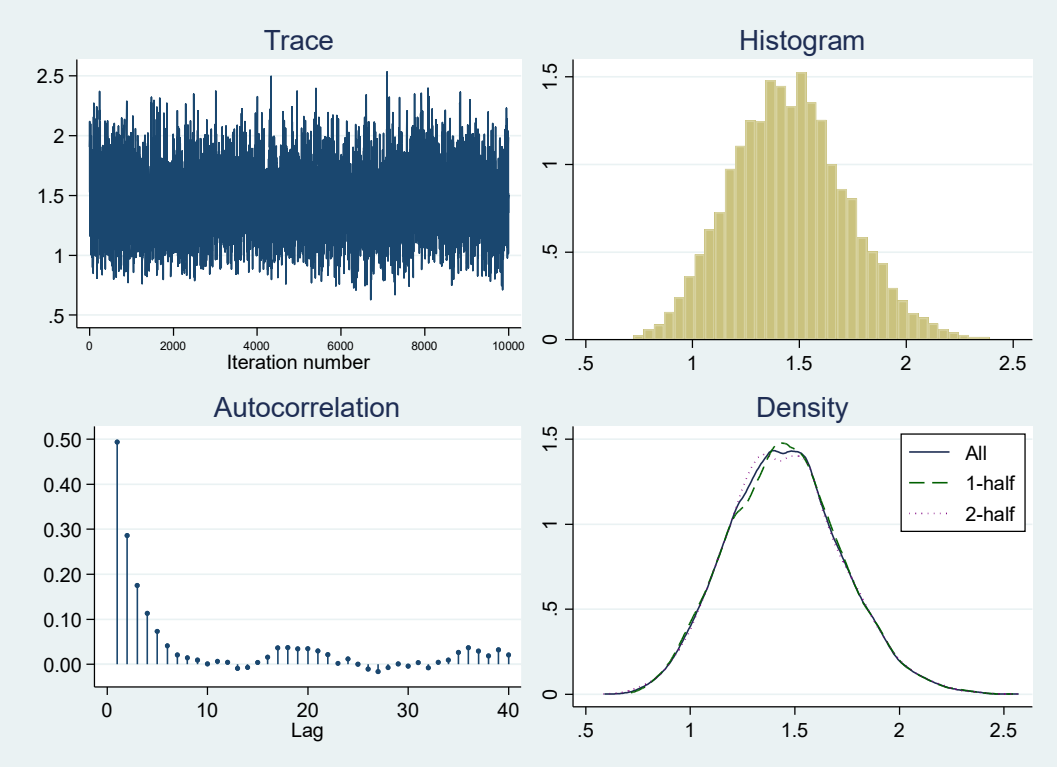


Figure C8. Posterior diagnostic plot for f

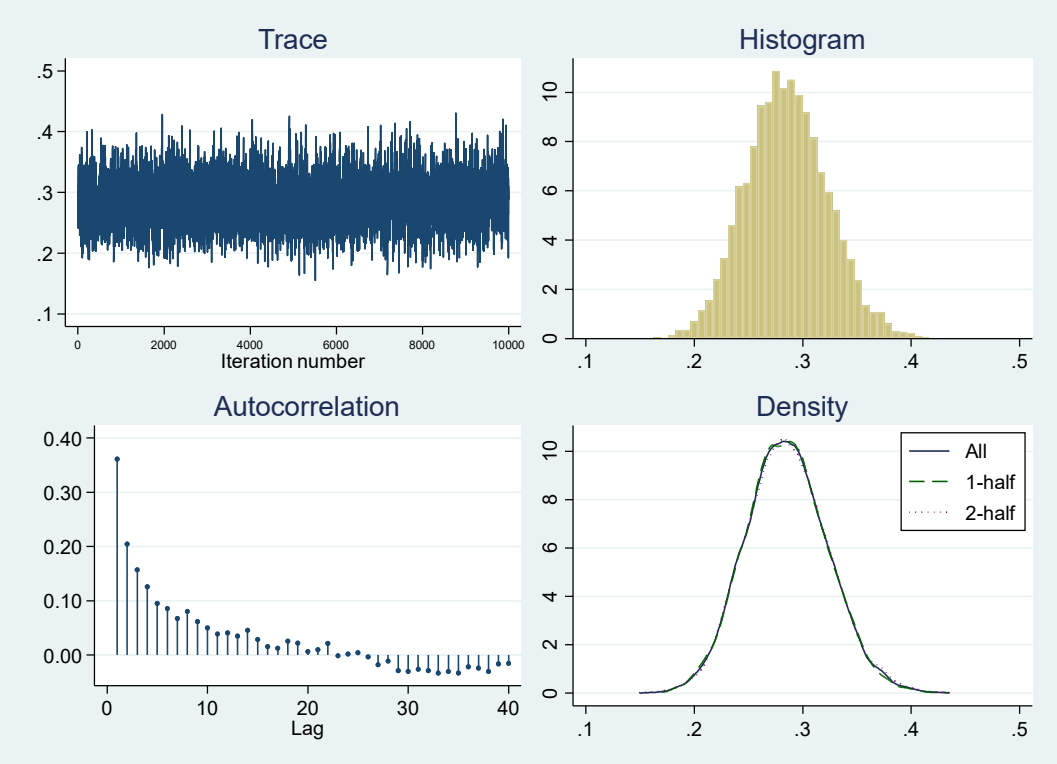


Figure C9. Posterior diagnostic plot for κ_K

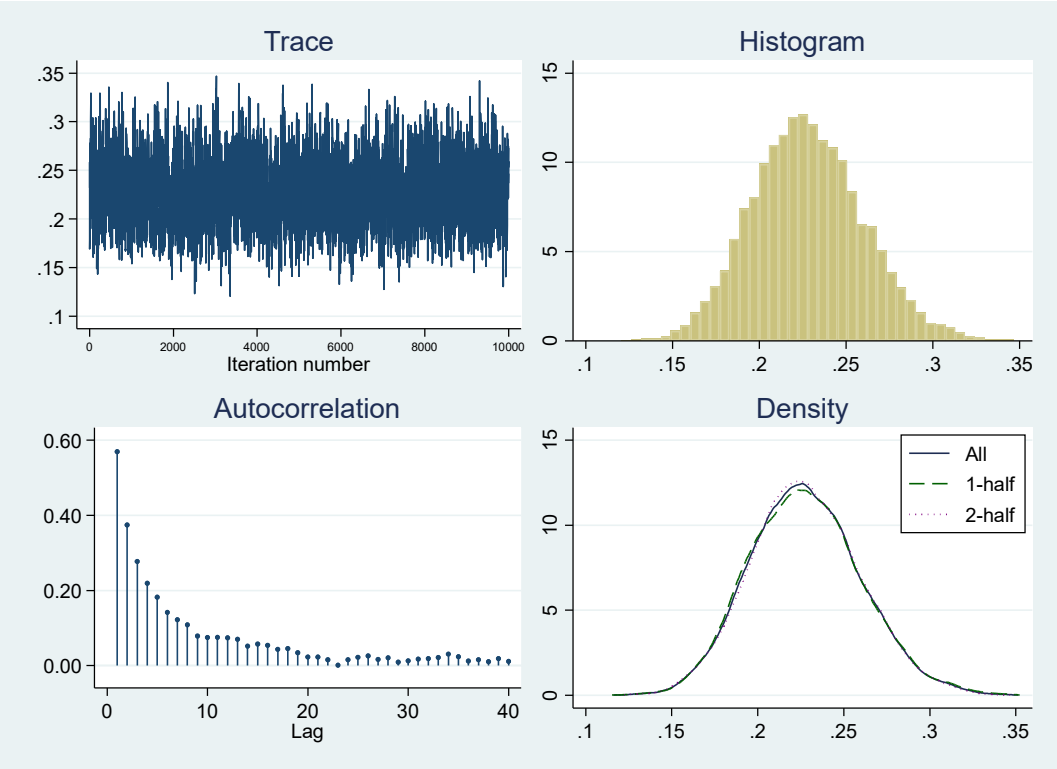
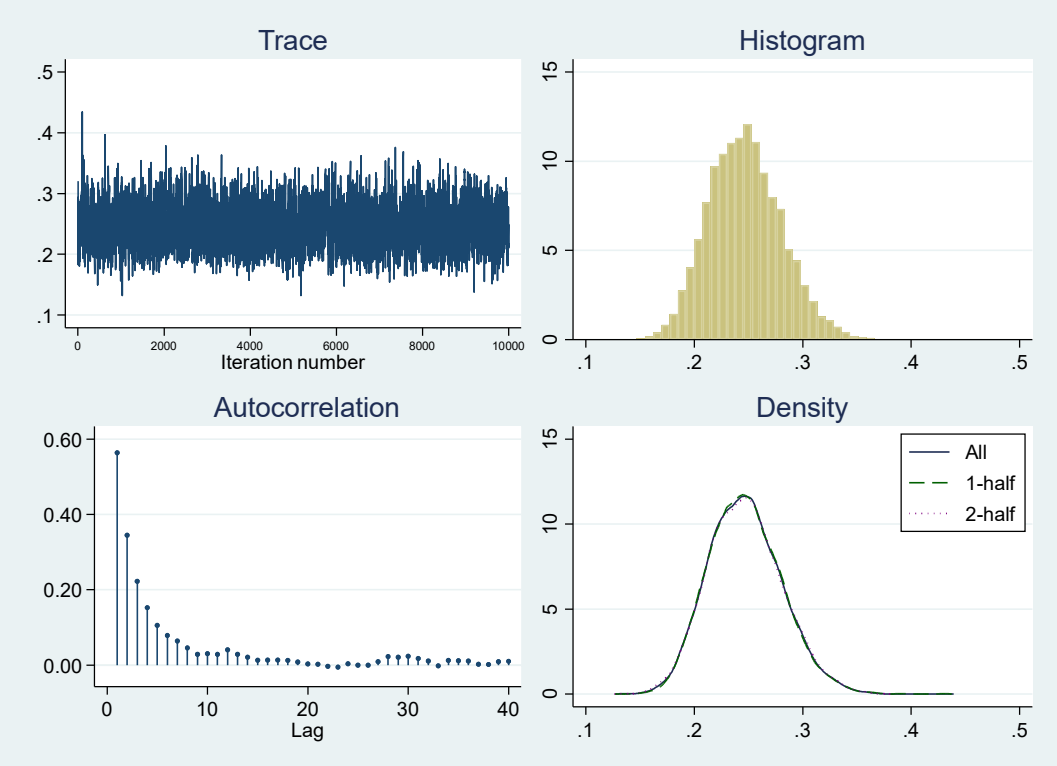


Figure C10. Posterior diagnostic plot for κ_E



Appendix C2: Representation of τ_S under Energy-Structure Normalization

In Equation (12), the term Ω_3 captures the effects of manufacturing emissions intensity and carbon abatement through CCS. On this basis, we specify $\tilde{\phi}_F$ in Equation (9) as:

$$\tilde{\phi}_F = \phi_F(b_{EF}a_{YE} + a_{YF})$$

where ϕ_F denotes the CO_2 emission factor of fossil-fuel use, b_{EF} the input-output coefficient of fossil fuels in electricity production, and $a_{YE} = \frac{E^Y}{Y}$ and $a_{YF} = \frac{F^Y}{Y}$ are input-output coefficients of electricity and fossil fuels in manufacturing production.

To ensure analytical tractability, we impose the following assumptions.

Assumption C2-1. As an empirical restriction, the input-output coefficients a_{YE} and a_{YF} are assumed to be constant over time.

Assumption C2-2. The composite emission intensity of electricity generation, $\phi_F b_{EF}$, is constant over time.

Assumption C2-1 captures the empirically prevalent feature of the Leontief input-output framework, characterized by fixed technical coefficients (Australian Bureau of Statistics, 2021). This restriction does not redefine the coefficients as primitive technological constants; rather, it constrains the corresponding equilibrium coefficients to remain approximately stable over the sample period.

Assumption C2-2 abstracts from drastic energy transitions. Although energy transition is allowed in Remark 1(ii), a substantial transition is difficult under the path-dependent nature of energy technology adoption and stable emissions structure of the electricity sector discussed in **Appendix B2**. While Assumption C2-2 focuses on medium-run persistence in emission intensity driven by technological and infrastructure constraints, this persistence is consistent with the stable emissions structure of the electricity sector documented in **Appendix B2**.

Together, Assumptions C2-1 and C2-2 ensure that the composite term $\tilde{\phi}_F$ is time-invariant. We then normalize $\tilde{\phi}_F = 1$ as a scale convention to identify τ_s from the composite parameter Ω_3. Accordingly, τ_s is interpreted as a standardized CCS-equivalent mitigation parameter rather than a directly measured physical capture rate.

Under this normalization, the emission component in Equation (12) can be rewritten as:

$$\Omega_3 Y = (1 - \tau_s) Y$$

which directly implies:

$$\tau_s = 1 - \Omega_3.$$

References for Appendix C2

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Appendix C3: Carbon Capture Requirements and Natural Carbon Absorption Mechanisms

The decarbonization process underlying sustainability in this model is characterized by τ_s and τ_{nx}. These parameters are structurally recovered from the estimated DSGE coefficients. Although the CO_2 stock is treated as predetermined, the model allows for endogenous mitigation adjustments through changes in CCS deployment and vegetation restoration (denoted by ΔCCS and $\Delta vegetation$), while treating the structural parameters τ_s and τ_{nx} as given. Under this framework, adjustments in mitigation efforts imply a substitution margin between CCS deployment and vegetation restoration: greater reliance on vegetation restoration can reduce the amount of CCS required to achieve a given emissions-reduction target, and vice versa. Such adjustments can be interpreted as mitigation choices associated with production activities, allowing the economy to achieve sustainability without altering the underlying economic structure. In this sense, the estimated parameters characterize the baseline carbon dynamics, whereas mitigation adjustments represent policy margins available within the existing economic structure.

We compute τ_{nx} according to Equation (12-2) as $\tau_{nx} = \Omega_4 - \tau_{n0}$, where τ_{n0} is the sample mean of the observed values reported in Table B1 of **Appendix B1**. We derive τ_s as $1 - \Omega_3$ from Equation (12-1) by normalizing $\tilde{\phi}_F$ to unity (see **Appendix C2** for details). Equations (12-1) and (12-2) constitute a one-to-one mapping between the estimated parameters (Ω_3, Ω_4) and the structural parameters (τ_s, τ_{nx}), while the remaining parameters Ω_0, Ω_1, and Ω_2 are irrelevant for this transformation. Thus, τ_s and τ_{nx} are uniquely identified within the macroeconomic block.

We report τ_s and τ_{nx} in Panel B of Table B2. Under the adopted normalization, the estimated value of τ_s=0.7144 represents a standardized CCS-equivalent mitigation parameter,

implying that the baseline trajectory requires a modeled removal flow equivalent to 71.44% of the standardized emissions flow. While CCS can achieve high capture rates (Metz et al., 2005), its large-scale deployment has been constrained in part by significant and heterogeneous costs (Baylin-Stern and Berghout, 2021). Thus, within the model's mitigation structure, vegetation restoration can serve as a complementary measure to address the remaining emissions gap.

The vegetation restoration parameter $\tau_{nx} = -0.0828$ indicates a negative contribution of human activities to vegetation coverage. When normalized by the baseline absorption coefficient, τ_{n0}, reported in Table B1 (**Appendix B1**), this corresponds to an estimated decline in carbon sequestration capacity of approximately 45.77% (≈ 0.0828 ÷ 0.1809), conditional on the primitive structure of the current growth trajectory. This estimate is of a similar order of magnitude to the 36% reduction in carbon sink capacity reported for Northern Hemisphere boreal forests by Shchepashchenko et al. (2024), although the two estimates refer to different geographic and ecological scopes and are therefore not directly comparable. The estimated deterioration in carbon absorption capacity therefore highlights a potential sustainability concern within the model's baseline trajectory.

Taken together, the results indicate that relying heavily on CCS would entail a substantial mitigation requirement, while the literature documents significant costs associated with large-scale CCS deployment. At the same time, the negative contribution of human activities to natural carbon absorption indicates a potential margin for improving vegetation-based sequestration. Enhancing nature-based carbon sequestration could provide an additional source of CO_2 removal, complement CCS, and contribute to a more diversified mitigation strategy.

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Baylin-Stern, A. and N. Berghout (2021) *Is carbon capture too expensive?* International Energy Agency (IEA).

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Shchepashchenko, D., Shvidenko, A., & Heyl, A. (2024) *Forests endure as carbon sink despite regional pressures*. International Institute for Applied Systems Analysis (IIASA).

Appendix D: Policy Analysis and Carbon-Sequestration Strategies

Appendix D1: Policy Counterfactual Analysis of the Effects of Carbon Taxation on Consumption, Output, and the CO_2 Stock

Using the estimates from the DSGE model, this section presents a reduced-form carbon-tax policy counterfactual to examine its effects on consumption, output, and the CO_2 stock within the macroeconomic block. Following the counterfactual approach of Pesaran and Smith (2018), policy effects are evaluated by comparing two alternative policy regimes: a regime without a carbon tax and a regime with a carbon tax:

System 1:

$$D_{t+1} = \Omega_3 Y_t - (\Omega_4 - 1)D_t + \epsilon_t^D,\ \epsilon_t^D \sim N(0,\sigma_D^2) \quad \text{(D-1-1)}$$

$$C_{t+1} = \Omega_0 C_t + \epsilon_t^C,\ \epsilon_t^C \sim N(0,\sigma_C^2) \quad \text{(D-1-2)}$$

$$Y_{t+1} = \frac{C_{t+1}+\Omega_2 D_{t+1}}{\Omega_1} \quad \text{(D-1-3)}$$

System 2:

$$D_{t+1} = \Omega_3 Y_t - \big((1+\mathrm{t_D})\Omega_4 - 1\big)D_t + \epsilon_t^D,\ \epsilon_t^D \sim N(0,\sigma_D^2) \quad \text{(D-2-1)}$$

$$C_{t+1} = (1-\mathrm{t_D})\Omega_0 C_t + \epsilon_t^C,\ \epsilon_t^C \sim N(0,\sigma_C^2) \quad \text{(D-2-2)}$$

$$Y_{t+1} = \frac{C_{t+1}+\Omega_2 D_{t+1}}{\Omega_1} \quad \text{(D-2-3)}$$

For the counterfactual exercise, we impose a proportional policy wedge on the estimated system. The policy intervention is implemented by modifying the carbon accumulation parameter and introducing a proportional consumption wedge associated with the carbon tax. The resulting equilibrium path of output is then recovered using the estimated equilibrium relationship. Following Pesaran and Smith (2018), the same realizations of the structural shocks are imposed under both policy regimes so that differences in the simulated paths can be attributed solely to the policy intervention. The effect of a carbon tax $\mathrm{t_D}$ is reflected in the reduction of the CO_2 stock through Equation (D-2-1). The term $\mathrm{t_D}$ is interpreted as a

proportional policy wedge intended to capture the Pigouvian response to the CO_2 externality, as specified in Equation (D-2-2). Moreover, Equations (D-1-2) and (D-2-2) are recursive approximations to the consumption dynamics implied by the estimated Euler equation (10); Equations (D-1-1) and (D-2-1) are based on Equation (12); Equations (D-1-3) and (D-2-3) are based on Equation (11'). These counterfactual equations are intended as a reduced-form policy representation of the estimated DSGE dynamics rather than as a re-solved equilibrium under an explicitly specified tax-inclusive utility and budget structure.

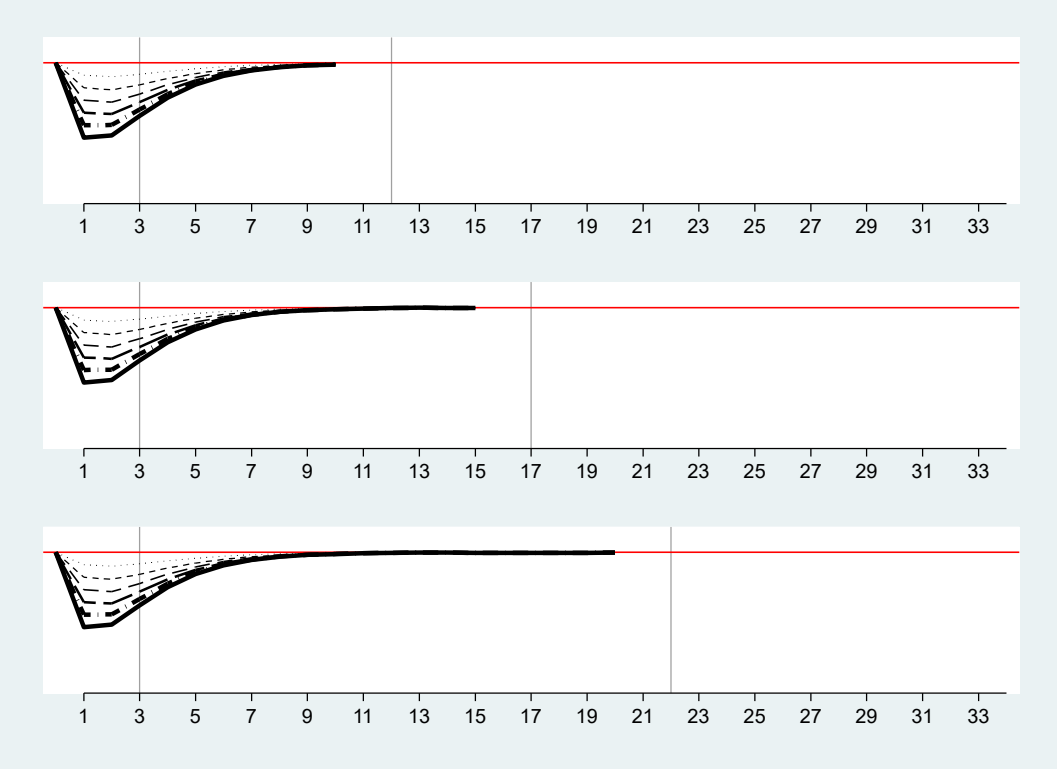
The reduced-form policy counterfactuals are conducted using the estimated structural parameters reported in Table B2. The structural shocks are assumed to be independently and identically distributed normal random variables with zero means and fixed variances throughout the counterfactual exercises. For simplicity, the baseline reduced-form policy counterfactuals set $\sigma_D^2 = \sigma_C^2 = 0.01$ as fixed values for the counterfactual simulations. The counterfactual policy exercises are conducted under different tax rates and implementation horizons. Six tax rates (5%, 10%, 15%, 20%, 25%, and 30%) are considered, each under three policy implementation durations (10, 15, and 20 years), over the period 1990–2023.

Comparisons between Systems 1 and 2 yield the policy counterfactual effects illustrated in Figures D1-D3. Under the specified counterfactual policy system, higher carbon-tax rates reduce simulated consumption, output, and the CO_2 stock. In addition, extending the policy implementation horizon produces more persistent simulated effects on all three variables.

References for Appendix D1

Pesaran, M. Hashem & Ron P. Smith (2018) Tests of policy interventions in DSGE models. *Oxford Bulletin of Economics and Statistics* 80(3), 457-484.

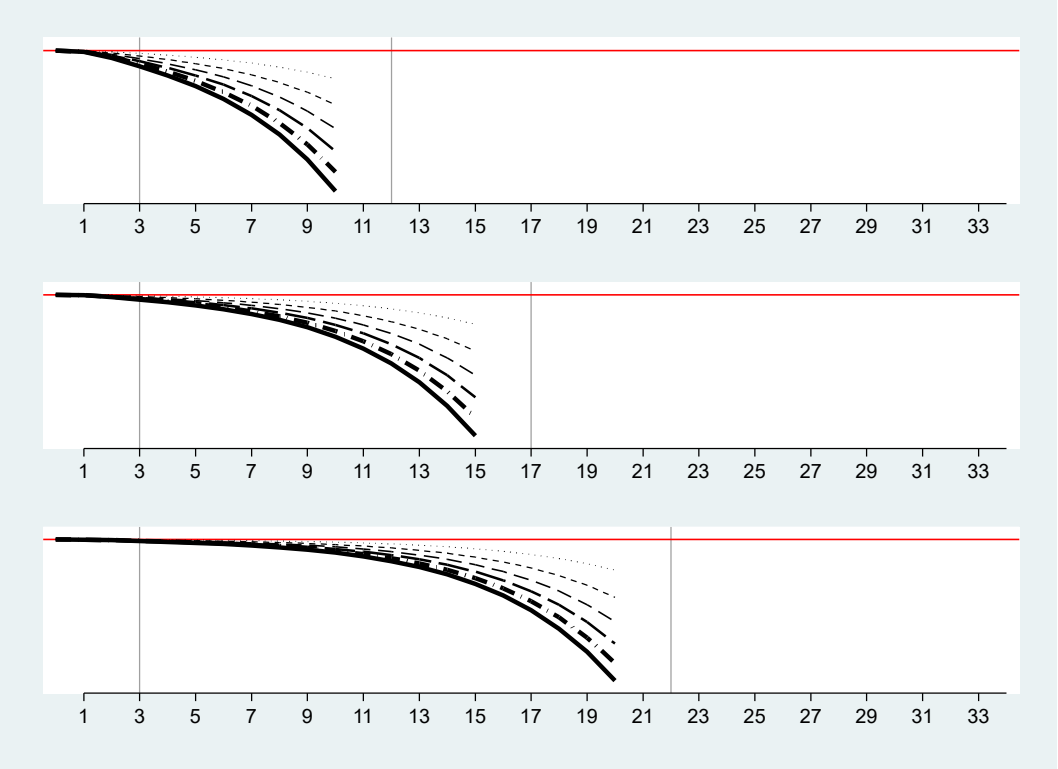
Figure D1. Impacts of a carbon tax on consumption



Note:

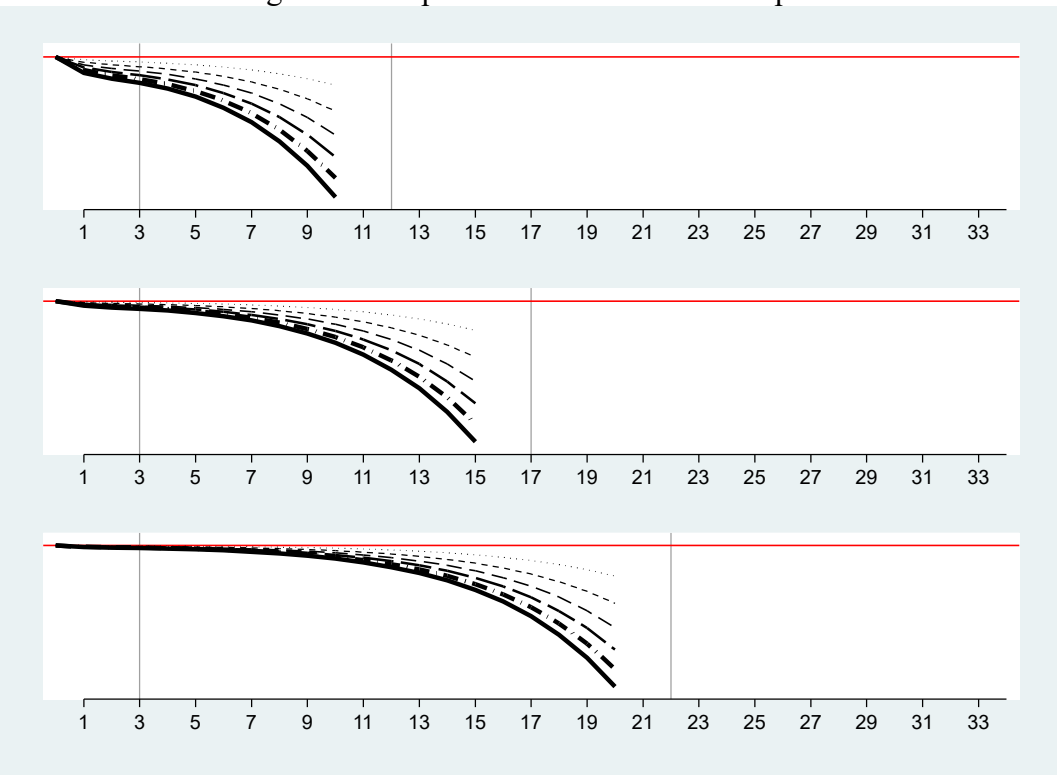
Six tax rates (5%, 10%, 15%, 20%, 25%, and 30%) and three implementation horizons (10, 15, and 20 years) are considered within the 34-period baseline window of the reduced-form policy counterfactuals, covering the period 1990–2023. The six tax rates are distinguished by different line styles, ranging from a thin dotted line to a thick solid line. The three implementation horizons are indicated by the vertical lines shown in the three panels of the figure.

Figure D2. Impacts of a carbon tax on CO_2 stock



Note: See note for Figure D1.

Figure D3. Impacts of a carbon tax on output



Note: See note for Figure D1.

Appendix D2. Desert Greening as a Complementary Carbon-Sequestration Strategy

Given the previously discussed mitigation requirements associated with CCS and the economic implications of carbon taxation, it is useful to explore vegetation restoration as a complementary source of CO_2 removal. Nevertheless, although some U.S. cities have adopted ambitious targets to increase tree cover by 50% to 100% (e.g., Hilbert et al., 2019; Leff, 2016), the high cost of urban land remains a concern. Against this backdrop, desert greening has been proposed as a potential approach to vegetation restoration for CO_2 removal.

First, recent studies indicate that rising atmospheric CO_2 levels have enhanced global plant photosynthesis, thereby supporting desert greening. This phenomenon—known as the CO_2 fertilization effect—has benefited approximately 41% of the world's drylands, contributing to improved vegetation cover and reduced desertification (Pearce, 2024). Moreover, increased rainfall in regions such as the Sahara (NASA, 2024), the Middle East (Li et al., 2023), and desert areas in the southwestern United States (UC Riverside, 2022) has created new opportunities for large-scale desert greening.

Furthermore, compared to urban tree-planting expenses, international experience suggests that large-scale dryland restoration can be implemented at costs that may be lower than those associated with urban tree planting. According to China's Green Great Wall program, greening 9 million acres of arid land costs approximately $8 billion (Hashmi, 2021), or about $889 per acre. Therefore, although desert greening involves various considerations, such as the opportunity cost of land, irrigation and maintenance costs, and vegetation survival rates, its potential advantages remain significant.

We model desert greening ($\Delta\tau_{nx}$) as a fraction z_{nx} of the natural carbon absorption rate (τ_{n0}), i.e., $\Delta\tau_{nx} = z_{nx}\tau_{n0}$. For tractability, we adopt a linear first-order approximation over the

range of restoration considered. This policy counterfactual is motivated by the relatively low implementation cost per acre and potential synergies with large-scale solar deployment. The benefits of solar deployment extend beyond power generation. In China, photovoltaic panels installed in arid regions may facilitate vegetation growth by modifying local microclimatic conditions, such as reducing wind speed and surface evaporation (NASA Earth Observatory, 2024). Nevertheless, this specification is not intended to imply constant marginal returns at larger scales, where saturation effects and ecological nonlinearities may become important.

On this basis, the resulting change in the CO_2 stock is expressed as $\Delta D(z_{nx} > 0) = \Omega_3 Y - \big((\tau_{nx} + \Delta\tau_{nx}) + \tau_{n0}\big)D$, where $\Delta\tau_{nx} = z_{nx}\tau_{n0}$, and both Y and D are standardized. Comparing this expression with the corresponding change in CO_2 stock without desert greening allows us to isolate the incremental contribution of desert greening to carbon reduction.

To capture the extent to which desert greening can contribute to CO_2 reduction relative to CCS under the linear specification above, we define the relative carbon-reduction measure as $\frac{\Delta D(z_{nx}>0)-\Delta D(z_{nx}=0)}{\tau_S Y}$. We simulate this measure at five levels of z_{nx} (10%, 20%, 30%, 40%, and 50%) over four periods (2003–2007, 2008–2012, 2013–2017, and 2018–2022). A more negative value indicates a larger incremental reduction in the CO_2 stock relative to the baseline carbon-capture flow associated with CCS.

The results reported in Figure D4 show that the relative carbon-reduction contribution of desert greening increases monotonically with z_{nx} across the four periods. The estimated changes range from approximately −3% to −7% at z_{nx}=10% and from approximately −14% to −37% at z_{nx}=50%. Holding other factors constant, these results imply that, under the specified linear adjustment to the natural absorption coefficient, the incremental reduction in the modeled carbon stock at z_{nx}=10% is equivalent in magnitude to approximately 3%–7% of the standardized baseline CCS carbon-capture flow across the four periods. At z_{nx}=50%, the corresponding magnitude ranges from approximately 14% to 37% of the standardized baseline

CCS flow. These results illustrate the potential contribution of desert greening as a complementary source of CO_2 removal alongside existing carbon-sequestration strategies, under the model's linear approximation.

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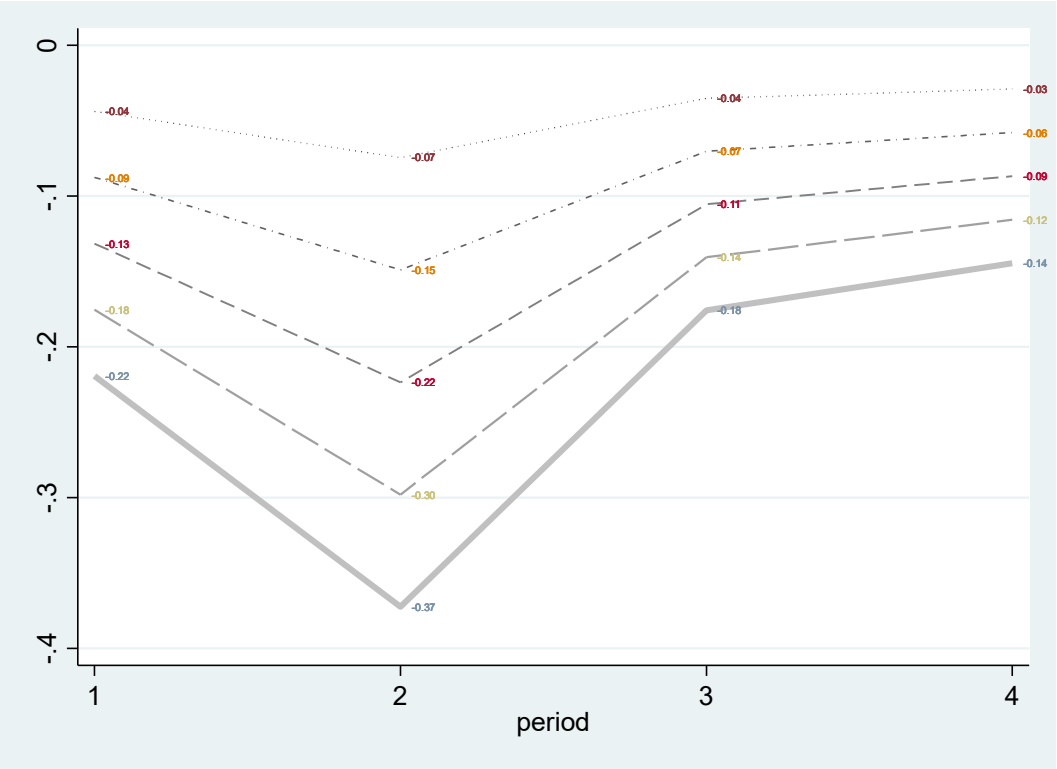
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UC Riverside (2022) What Does the Summer Rain Do? *Latest Cal Nat Blog*. Palm Desert Center.

Figure D4. Carbon-reduction contribution of desert greening over time



Notes:

(i) The relative carbon-reduction contribution of desert greening is measured by normalizing its incremental impact on the CO_2 stock by the baseline CCS carbon-capture flow, given by the product of the CCS carbon-capture rate (τ_s) and economic output (Y): $\frac{\Delta D(z_{nx}>0)-\Delta D(z_{nx}=0)}{\tau_s Y}$. This measure indicates the relative contribution of desert greening to carbon reduction compared with carbon capture via CCS.

(ii) We compute the average relative carbon-reduction contribution of desert greening for each z_{nx} level (10%, 20%, 30%, 40%, and 50%) over four periods: 2003–2007 (Period 1), 2008–2012 (Period 2), 2013–2017 (Period 3), and 2018–2022 (Period 4). These periods are indexed as Periods 1–4 on the horizontal axis, while the corresponding average relative carbon-reduction contributions are shown on the vertical axis. In the figure, the five z_{nx} levels are represented by different line styles: 10% by a thin dotted line, 20% by a thin short dash-dot line, 30% by a medium thin dashed line, 40% by a medium long-dashed line, and 50% by a thick solid line.